Uncertainty Quantification for Super-Resolution with Diffusion Priors

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ABSTRACT

Diffusion-based priors achieve strong performance on image inverse problems, but it is unclear whether the stochastic reconstructions they produce behave like calibrated Bayesian posterior samples. We study uncertainty quantification for bicubic super-resolution on ImageNet using DDRM and the posterior targeting baseline FPS-SMC. We consider scale factors $s \in \{4, 8\}$ and noise levels $\sigma_0 \in \{0, 0.05\}$, and for each test image we fix the measurement y and draw multiple reconstructions conditioned on the same y. We evaluate accuracy of the sample mean and uncertainty behaviour using marginal calibration, sharpness, usefulness, and frequency diagnostics. Both solvers exhibit strong marginal under coverage: at nominal level 0.95, DDRM achieves coverage around 0.47 to 0.53 across settings, while FPS-SMC drops to around 0.35 to 0.36 at $s = 8$ with $\sigma_0 = 0.05$ under the tested particle budgets. Despite miscalibration, variance maps remain informative for error localisation, and DDRM shows higher Spearman correlation between pixelwise error and variance in the noisy settings. Frequency diagnostics show underrepresented high-frequency uncertainty, with an under-exploration ratio $R_{\mathrm{UE}} < 1$ across all settings. These results support the accuracy trap perspective and show that uncertainty must be evaluated directly when using diffusion priors for super-resolution.

1 Introduction

Super resolution (SR) aims to recover a high-resolution (HR) image from a low-resolution (LR) observation. This is an ill-posed inverse problem, since many possible HR images can match the same measurement [1, 2]. A common controlled setting uses bicubic downsampling, which is often adopted in SR benchmarks to study solver behaviour under known degradations [3, 4].

Diffusion models provide a strong learned prior for natural images [5, 6]. They have recently been used as plug-and-play (PnP) priors for inverse problems, in which a pretrained generative model is combined with a task-specific data-consistency term [7, 8]. Several diffusion-based solvers have been proposed for image restoration and inverse problems, including DDRM and related posterior sampling approaches [9, 10, 11, 12, 13]. These methods are stochastic, which means that for a fixed measurement y, repeated runs can produce different reconstructions. This stochasticity is often interpreted as uncertainty quantification (UQ), but it is not guaranteed to correspond to a calibrated Bayesian posterior.

Most SR evaluations focus on point estimate metrics such as PSNR and SSIM. However, good point accuracy does not imply reliable uncertainty. Qiu et al. [14] refer to this as the accuracy trap, which means that two methods can achieve similar point metrics while producing uncertainty estimates with different calibration and sharpness behaviour. This motivates evaluating uncertainty explicitly using calibration style diagnostics [15, 16, 17].

In this paper, we evaluate uncertainty for diffusion-based SR solvers using a benchmark based on calibration, sharpness, usefulness, and frequency diagnostics. Our main solver is DDRM [9], and it is compared against FPS-SMC, which targets the posterior for linear inverse problems through a filtering and sequential Monte Carlo perspective [18]. We follow the recent benchmarking paper of Qiu et al. [14], and focus on whether the induced sample distribution behaves like a calibrated posterior sampler for SR.

We study bicubic downsampling with scale factor $s \in \{4, 8\}$ and additive Gaussian noise with standard deviation $\sigma_0 \in \{0, 0.05\}$. For each test image, we fix a measurement y and draw multiple reconstructions conditioned on the same y. This enables empirical evaluation of conditional uncertainty. We report the accuracy of the sample mean and evaluate uncertainty using the framework in Section 3, including frequency diagnostics tailored to SR.

Our contributions are as follows. First, we provide an empirical uncertainty benchmark for diffusion-based SR under controlled bicubic degradations and a fixed compute budget. Second, we compare DDRM with a posterior targeting baseline, FPS-SMC, under a shared evaluation protocol. Third, we include frequency-based diagnostics that test whether uncertainty is allocated to high-frequency content that is difficult to recover from the LR measurement.

The remainder of the dissertation is organised as follows. Section 2 introduces SR as a linear inverse problem and reviews diffusion priors for inverse problems. Section 3 defines the uncertainty benchmarking framework. Section 4 presents DDRM and FPS-SMC. Section 5 describes the experimental protocol. Section 6 reports results and analysis.

2 Background & Related Work

2.1 SR as a Linear Inverse Problem

Super resolution (SR) can be viewed as an image inverse/reconstruction problem, that is, we observe a degraded measurement y and aim to recover an unknown clean high resolution (HR) image x. This low-resolution (LR) observation y is generated from the HR image x through a known degradation process, i.e. it can be written as

$$y = A_s(x) + \epsilon, \tag{1}$$

where A_s is a known linear forward operator associated with scale factor s, and ϵ denotes measurement noise.

In this report, A_s corresponds to bicubic downsampling by an integer factor s. Concretely, A_s applies a fixed bicubic filtering operation and then subsamples to produce an LR image. This choice provides a controlled and widely used benchmark setting for studying SR solvers [3, 4].

SR problem is ill-posed because the downsampling operation removes a lot of information. In particular, high-frequency components of x are diminished or removed by the blur and downsampling process, so many distinct HR images can essentially map to the same LR measurement. This non-uniqueness is not only a theoretical issue, as in natural images, fine textures and sharp edges often lie in frequency bands that are weakly constrained by the measurement, which means that plausible reconstructions can differ substantially while still being consistent with y.

A convenient way to understand this ill-posedness is using the idea of observable versus null/missing components. For linear operator A, one can decompose the image space into directions that influence the measurement and directions that do not. This viewpoint is central to diffusion-based solvers such as DDRM, which explicitly separates measurement-constrained directions from the directions that are purely governed by the learned image prior [9]. For the SR setting, this decomposition aligns well with a frequency domain intuition, that is, low-frequency content is relatively well constrained by the LR observation, while the missing high-frequency details behave like a null space, or missing parts, that must be inferred from prior knowledge rather than directly observed data.

In this paper, we focus on SR with bicubic downsampling at scale factors $s \in \{4, 8\}$, with optional additive Gaussian noise. And the goal is not only to produce a single reconstruction, but to characterise uncertainty over plausible HR images that are consistent with the same measurement y.

2.2 Bayesian View of SR and Uncertainty

A Bayesian formulation makes the role of uncertainty explicit. The key idea is to represent (i) how measurements are generated from an unknown image, and (ii) what images are considered to be plausible, using a likelihood and prior. Given the forward model $y = A(x) + \epsilon$, Bayesian inference represents uncertainty about x after observing y using the posterior distribution

$$p(x|y) \propto p(y|x)p(x), \tag{2}$$

where $p(y|x)$ encourages consistency with the observation model and $p(x)$ encodes prior knowledge about natural images. For many restoration tasks, including SR, a common likelihood model assumes additive Gaussian noise $\epsilon \sim \mathcal{N}(0, \sigma^2 I)$, which gives

$$\log p(y \mid x) = -\frac{1}{2\sigma^2}\|y - A(x)\|^2 + \text{const.} \tag{3}$$

This likelihood model is standard in diffusion-based restoration methods for linear inverse problems, including DDRM [9].

Ideally, we need a maximised posterior likelihood, so substituting into the posterior 2 gives the standard maximum a posteriori (MAP) objective

$$\hat{x}_{\text{MAP}} = \arg\max_x \log p(x \mid y) = \arg\min_x \frac{1}{2\sigma^2}\|y - A(x)\|^2 - \log p(x). \tag{4}$$

This formulation has been widely used in many PnP-based solvers [7], and can be viewed as a trade-off between data consistency and prior regularization.

However, MAP provides only a single point estimate, while uncertainty quantification requires access to the distribution $p(x|y)$. In ill-posed inverse problems such as SR, posterior uncertainty is fundamental; even when measurement noise is small, the posterior may remain broad along directions that are weakly constrained by the operator A, for example, missing high-frequency content in SR. The UQ benchmarking study of plug and play diffusion priors emphasises that inverse problems are inherently distributional, and that evaluating only a single reconstruction using metrics

such as PSNR can be misleading when a solver induces a non-trivial distribution over outputs [14]. This motivates evaluating sampling-based, specifically diffusion-based SR solvers using distributional criteria rather than relying solely on point-estimate accuracy.

In this work, we evaluate whether repeated stochastic runs of a diffusion-based model behave like samples from a fixed posterior $p(x|y)$ for a given measurement y. In other words, when measurement noise is present, it is important to hold y fixed across runs. Otherwise, repeated reconstructions correspond to sampling from different posterior $p(x|y)$ and cannot be meaningfully evaluated using calibration metrics such as coverage. This fixed measurement protocol allows us to interpret the empirical sample distribution induced by the solver as an approximation to the conditional distribution over HR images.

Finally, a calibrated Bayesian posterior has precise operational meaning. For instance, for a marginal $(1-\alpha)$ credible interval derived from $p(x \mid y)$, the true image x should fall inside the interval with probability approximately $(1-\alpha)$ under the data-generating process. In later sections, we will test this property empirically using coverage-based metrics and reliability curves/calibration curves, which plot empirical coverage of nominal $(1-\alpha)$ intervals against the nominal level, this diagnostic is standard for calibration assessment [15, 17].

2.3 Diffusion Models as Priors

Generative models as image priors. A generative model aims to learn the distribution of natural images. Let $x \in \mathbb{R}^{H\times W\times C}$ denote an HR image. Training images can be viewed as samples from an unknown data distribution $p_{\text{data}}(x)$. A generative model introduces a parameterised distribution $p_\theta(x)$ that approximates $p_{\text{data}}(x)$. After training, we can generate images by sampling from $p_\theta(x)$.

This is useful for Bayesian super-resolution. In Eq. (2), the prior $p(x)$ favours realistic HR images and resolves the ambiguity caused by downsampling. A trained generative model provides such a prior by setting $p(x) = p_\theta(x)$. In this project, we use a diffusion model prior that is trained on the ImageNet dataset. Figure 1 gives a high-level view of training and sampling in generative modelling.

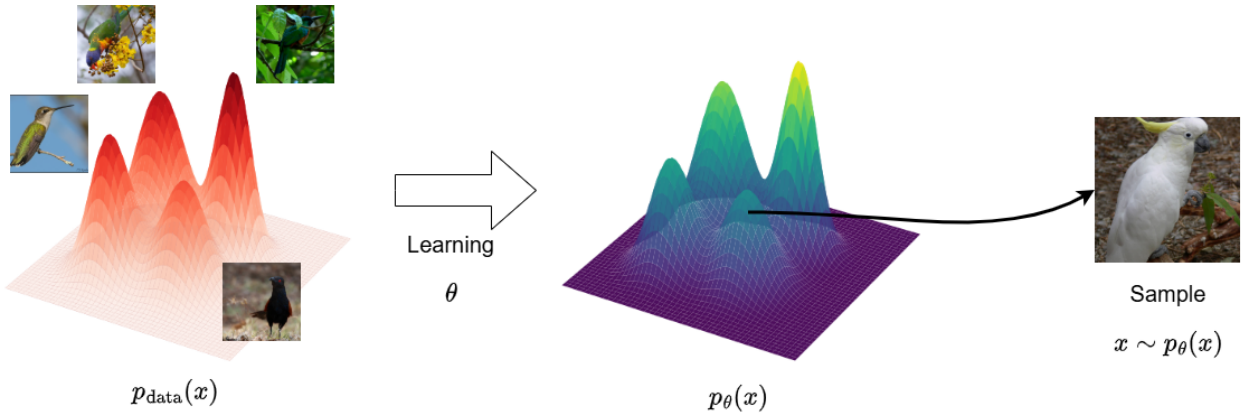


Figure 1: Generative modelling. Training data are samples from $p_{\text{data}}(x)$. Learning fits a model distribution $p_\theta(x)$. After training, we generate images by sampling $x \sim p_\theta(x)$. Figure created by the author.

Denoising diffusion probabilistic models. In this project, we describe the diffusion model using the denoising diffusion probabilistic model (DDPM) view, since it matches the pretrained diffusion prior used by DDRM and FPS [9, 5, 18]. A DDPM defines a forward process that gradually adds Gaussian noise to a clean image, and a learned reverse process that removes noise iteratively.

Let x_0 denote a clean image. The forward noising process produces latent states $x_1, ..., x_T$ using

$$q(x_t \mid x_{t-1}) = \mathcal{N}(\sqrt{\alpha_t}\, x_{t-1}, (1-\alpha_t)I)\,, \tag{5}$$

where $\{\alpha_t\}_{t=1}^T$ is a determined noise schedule with $\alpha_t \in (0,1)$. Due to the property of the Gaussian distribution, any state x_t has a closed-form representation using reparameterization:

$$x_t = \sqrt{\bar{\alpha}_t}\, x_0 + \sqrt{1-\bar{\alpha}_t}\, \epsilon, \qquad \epsilon \sim \mathcal{N}(0, I), \qquad \bar{\alpha}_t = \prod_{s=1}^{t} \alpha_s. \tag{6}$$

As t increases, the structure of the original image gradually fades away, and x_t becomes close to purely Gaussian noise $\mathcal{N}(0, I)$.

DDPM learns a reverse model that maps x_t back towards x_0. It defines Markovian transitions

$$p_\theta(x_{t-1} \mid x_t), \tag{7}$$

parameterised by a neural network with parameters θ. In common Gaussian parameterisations, each transition is Gaussian with a mean predicted by the network and a variance set by a schedule. A common choice is to predict the injected noise in Eq. (6). We denote this predictor by $\epsilon_\theta(x_t, t)$. It induces a clean image estimate

$$\hat{x}_0(x_t, t) = \frac{1}{\sqrt{\bar{\alpha}_t}}\Big(x_t - \sqrt{1-\bar{\alpha}_t}\,\epsilon_\theta(x_t, t)\Big). \tag{8}$$

The network output is then used to define the mean of $p_\theta(x_{t-1} \mid x_t)$. Other parameterisations predict a denoised image estimate directly, but the role is the same.

ELBO link and objective. DDPM is derived as a latent variable model with latent states $x_{1:T}$. A factorised variational distribution is introduced over the chain,

$$q(x_{1:T} \mid x_0) = q(x_T \mid x_0) \prod_{t=1}^{T} q(x_{t-1} \mid x_t, x_0), \tag{9}$$

which leads to an evidence lower bound (ELBO) on $\log p_\theta(x_0)$. Under the standard Gaussian construction, optimising this ELBO produces a simple denoising training objective. In particular, when the network is chosen to predict the injected noise, training is commonly implemented as

$$\min_\theta \; \mathbb{E}_{x_0,t,\epsilon}\left[\|\epsilon - \epsilon_\theta(x_t, t)\|^2\right], \tag{10}$$

where x_t is constructed from x_0 using Eq. (6) (detailed derivations are in [5]).

Sampling starts from $x_T \sim \mathcal{N}(0, I)$ and applies the learned reverse transitions repeatedly to draw a sample x_0 [5]. Figure 2 provides a simple view of the diffusion scheme, including a one-step reverse process. The current noisy state x_t is fed into the diffusion network, which produces a prediction. In DDPM, this prediction is often the injected noise $\hat{\epsilon}_\theta(x_t, t)$, but some parameterisations instead predict a denoised image estimate $\hat{x}_t$. The prediction is then used to compute the mean of the reverse transition $p_\theta(x_{t-1} \mid x_t)$. Finally, Gaussian noise is added to draw x_{t-1} from this distribution. This added noise keeps the reverse process stochastic, so repeated runs can produce diverse samples.

Score-based view. Diffusion models can also be described using a score-based view. At time step t, let $p_t(x)$ be the density of the noisy variable x_t. The score is $s_t(x) = \nabla_x \log p_t(x)$. The reverse time dynamics in the continuous time formulation depend on this score field, which provides a unified view of score-based models and diffusion probabilistic models [6].

A key practical question is how to obtain the score. For additive Gaussian noise, this can be addressed using Tweedie's formula that links the score of the noisy marginal to the posterior mean of the clean image. In particular, for $X_t = X_0 + \sigma_t Z$ with $Z \sim \mathcal{N}(0, I)$,

$$\nabla_{x_t} \log p_t(x_t) = \frac{\mathbb{E}[X_0 \mid X_t = x_t] - x_t}{\sigma_t^2}, \tag{11}$$

where σ_t is the noise standard deviation at time t. This relationship is stated and discussed in the inverse problem survey, and it is closely related to denoising score matching [8, 19, 20]. It shows that learning a denoiser that approximates $\mathbb{E}[X_0 \mid X_t = x_t]$ also gives an estimate of the score.

For DDPM-style models, the forward process is Gaussian and is often parameterised through a noise prediction network $\epsilon_\theta(x_t, t)$ [5]. This can be converted into an estimation of x_0 and therefore into a score estimate. This link explains why diffusion priors can be accessed through denoising or noise prediction outputs during sampling, rather than through explicit evaluation of the prior density $p_\theta(x)$. Finally, this score view is useful for inverse problems because the conditional score can be decomposed into prior score term and a measurement term:

$$\nabla_x \log p(x \mid y) = \nabla_x \log p(x) + \nabla_x \log p(y \mid x), \tag{12}$$

Diffusion-based inverse solvers can therefore also be viewed as combining a diffusion prior score with measurement information during sampling, although different methods implement this combination in different ways [14, 10, 21].

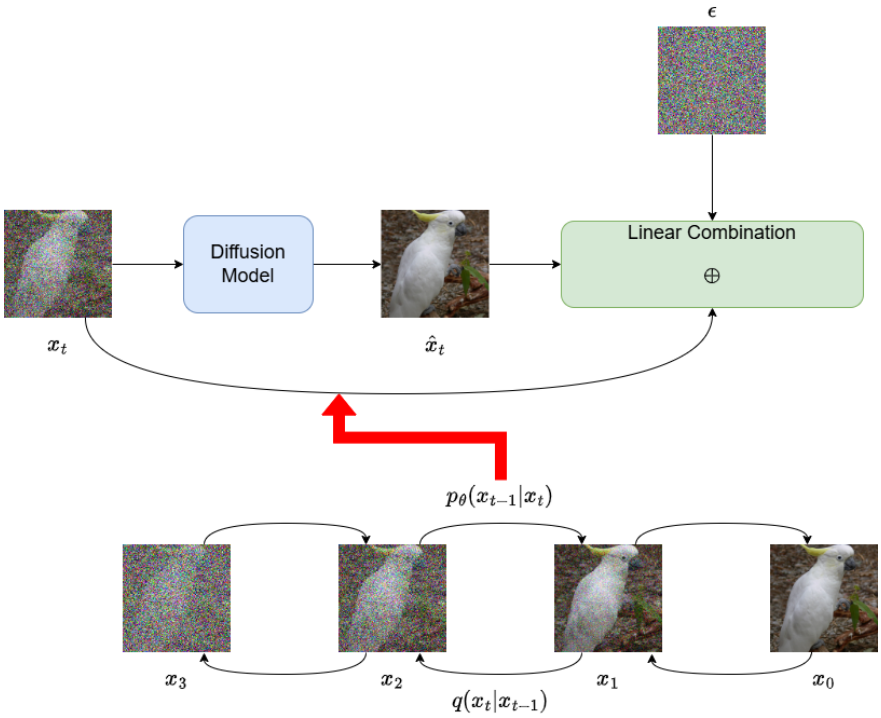


Figure 2: DDPM overview. The forward process gradually adds Gaussian noise to obtain x_t. The reverse process uses a neural network to define transitions $p_\theta(x_{t-1} \mid x_t)$ and iteratively produces a sample x_0. Figure created by the author.

2.4 Diffusion-Based Inverse Solvers for SR

Diffusion-based inverse solvers combine a pretrained diffusion prior with a known degradation model. In SR, we infer a HR image x from a LR measurement y under the forward model in Eq. (37). Using the diffusion prior $p_\theta(x)$ from Section 2.3, the Bayesian posterior can be reformulated as

$$p_\theta(x \mid y) \propto p(y \mid x)\, p_\theta(x). \tag{13}$$

The likelihood term $p(y \mid x)$ enforces measurement consistency. The prior term $p_\theta(x)$ encourages realistic natural images. However, exact sampling from Eq. (13) is intractable for high-dimensional data. So, practical methods modify the diffusion sampling process to produce results that are aligned with the measurement constraint.

Many recent methods use an unconditional diffusion model that is pretrained on clean images. This model defines an implicit prior distribution over images, but its likelihood $p_\theta(x)$ is not typically evaluated in closed-form. In practice, inverse solvers access the prior through the outputs of the diffusion neural network during the reverse process, such as noise prediction $\epsilon_\theta(x_t, t)$ or a denoised image estimate, which can also be converted into a score estimate $\nabla_{x_t} \log p(x)$. These outputs provide information at the current moment about what images are likely under the prior term and can be then combined with measurement consistency updates [19, 22, 23].

Existing solvers mainly differ in how they integrate the diffusion prior with the forward model. There are several surveys and benchmarking papers that organise diffusion-based inverse solvers into families. The survey by Daras et al. summarises methods and describes common ways to enforce measurement consistency, including gradients, projections, sampling, posterior approximations, and other optimisation-based methods [8]. InverseBench provides a benchmark view and groups many representative plug-and-play diffusion prior methods (PnPDP) into a small set of categories [22]. Here, we briefly explain the categories following InverseBench as below:

- **Linear guidance**. These methods exploit the linear structure of the forward operator and make consistency updates during sampling. They often require a pseudo-inverse or SVD property of A. DDRM is a representative method in this family and is designed for linear inverse problems [9]. InverseBench also lists related linear guidance methods such as DDNM and pseudoinverse guided diffusion models [22].

- **General guidance**. These methods introduce measurement information through guidance terms that are dependent on the likelihood. A common form uses a gradient of the likelihood/score with respect to the current iterate. DPS is a representative method in this family and is designed to handle general noisy inverse problems beyond simple linear settings [10].
- **Variable splitting**. These methods split inference into alternating steps. One step enforces measurement consistency using the forward operator. The other step applies the diffusion prior by running a short diffusion procedure. DiffPIR and PnP-DM are the representative examples listed in this category [11, 12].
- **Variational Bayes**. These methods introduce an explicit variational family to approximate the posterior and then optimise the variational parameters. InverseBench lists RED-diff as an example [13].
- **Sequential Monte Carlo**. These methods represent the posterior using a set of weighted particles. Particles are propagated along diffusion time steps, then reweighted using the likelihood, and resampled to focus on high probability regions. FPS is a representative method in this family and is designed to be asymptotically accurate as the number of particles increases, although practical performance depends on finite particle budgets [18].

For the scope of this paper, as super resolution with bicubic downsampling fits the linear operator setting, so linear guidance methods are natural candidates. We therefore use DDRM as the main solver [9]. We also consider FPS as a sequential Monte Carlo baseline in noisy settings where the likelihood is explicit [18]. FPS is also categorised as a posterior-targeting method in the UQ benchmarking taxonomy, which we will introduce in the next section, so this property motivates us to investigate the model compared to DDRM.

3 UQ Benchmarking Framework

This section defines how uncertainty is judged in this paper. It contains conceptual intuition with precise definitions of terminologies that we used across the study.

3.1 Why UQ must be evaluated

Diffusion-based inverse solvers are inherently stochastic. For a fixed measurement y, running multiple times can produce different reconstructions. This is an important property, since inverse problems are often ambiguous, and the Bayesian goal is a conditional distribution rather than a single image reconstruction. However, many papers only report a single reconstruction and focus on point estimate metrics such as PSNR or SSIM. The UQ benchmarking study argues that this can lead to a phenomenon called "accuracy trap". That means, two methods that achieve similar point metrics can produce very different uncertainty behaviour. This matters in many scientific and safety sensitive settings, such as medical imaging and where decisions depend on whether uncertainty is reliable [14].

Figure 3 provides an illustrative example of the accuracy trap for SR. Both solvers are applied to the same measurement. The posterior mean reconstructions have comparable PSNR, but the posterior variance maps differ in magnitude and spatial distribution. In this example, the DDRM variance is strongly concentrated around the object boundaries and fine edges. In contrast, FPS-SMC exhibits broader variance that is also visible across regions that are well constrained by the LR observation. This qualitative difference suggests that the solvers behave differently in terms of uncertainty, even when point estimate accuracy is close. This motivates evaluating uncertainty directly, rather than relying only on point estimate.

For this reason, this report not only evaluates accuracy but also assesses uncertainty directly. The goal is to test whether the solver induced distribution behaves like a calibrated approximation to the Bayesian posterior for SR.

3.2 UQ-based solver taxonomy

The UQ benchmarking study proposes a UQ-based categorisation of PnPDP [14]. The central distinction is the intended relationship between the solver-induced distribution and the Bayesian posterior. Below briefly explains the introduced UQ families:

Posterior targeting solvers. These methods are designed to approximate the Bayesian posterior distribution under stated assumptions. They often have a theoretical statement of correctness in an asymptotical sense.

Heuristic stochastic solvers. These methods produce variability due to the sampling procedure, but without a general posterior targeting guarantee. The induced distribution depends on design choices, such as conditional constructions, discretisation choices, and hyperparameters.

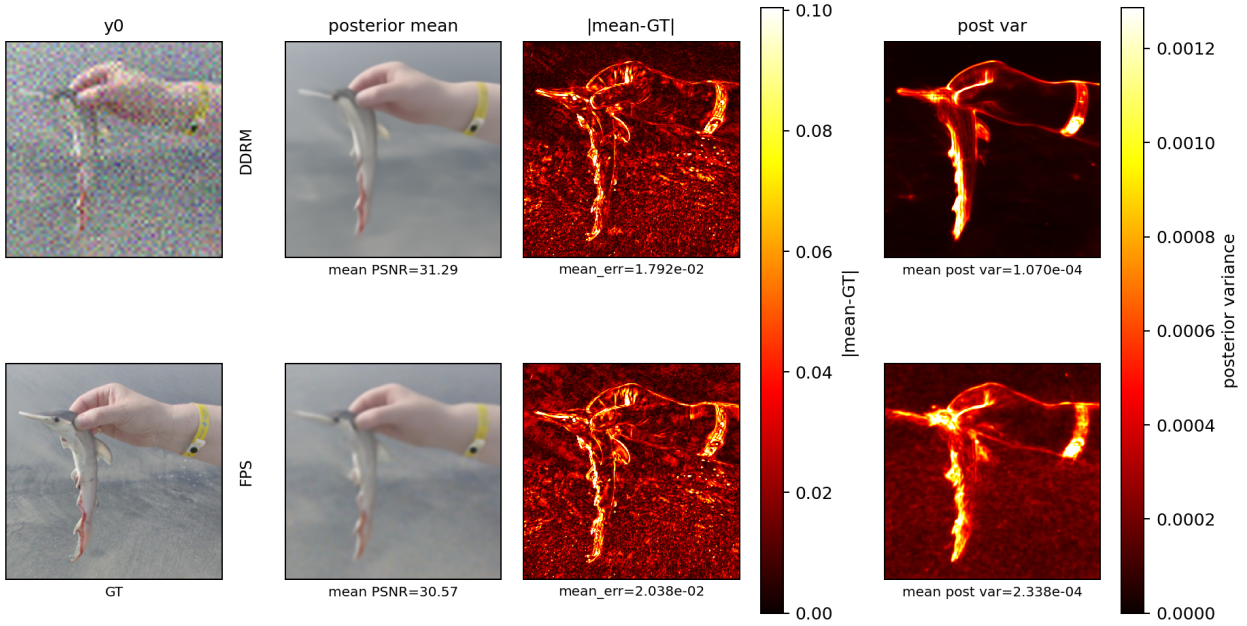


Figure 3: Illustrative example of the accuracy trap for super resolution. The same measurement is used for both solvers. We show the posterior mean, the absolute error of the posterior mean $|\hat{\mu} - x^*|$, and the posterior variance map $\hat{v}$. The posterior mean PSNR values are comparable, but the posterior variance maps differ in magnitude and spatial concentration.

MAP-like solvers. These methods behave mainly like point estimation and typically show limited diversity across runs.

This categorisation is different from structural taxonomies, such as the InverseBench categories in Section 2.4, which describe how a method is implemented and what access it needs to the forward model [22]. Both views are useful, structural taxonomies help decide whether a method can be applied to a given operator A. UQ-based taxonomies help decide what uncertainty behaviour one should expect, and what evaluation is needed.

In this work, DDRM is a linear guidance model in the structural taxonomy, while FPS is a sequential Monte Carlo method. In the UQ benchmarking taxonomy, FPS is discussed as a posterior-targeting method in an asymptotic sense, because sequential Monte Carlo targets a sequence of distributions and can converge as the number of particles increases [14, 18]. In contrast many guidance-based methods, such as DDRM, are discussed as heuristic, since their sampling distribution depends on design choices and does not come with a general theoretical posterior-targeting guarantee [14].

Overview of sampling and evaluation. For each ground truth image x^* and setting (s, σ_0), we form one measurement y using the bicubic operator A_s. When $\sigma_0 > 0$, we generate y once using a fixed measurement seed, then reuse the same y across all stochastic runs for that image. When $\sigma_0 = 0$, y is deterministic given x^* and A_s. We then run each solver multiple times with different run seeds, and pool the resulting reconstructions into a sample set. From this sample set, we compute the sample mean as a point estimate and the sample variance as a pixelwise uncertainty map. These outputs are evaluated using the accuracy and uncertainty metrics defined later in this section. Algorithm 1 summarises the full pipeline.

3.3 Sample-Based Uncertainty Representation

In this paper, uncertainty is defined through repeated solver outputs for a fixed measurement. For a given measurement y, we form a pooled sample set $\{x^{(k)}\}_{k=1}^{K'}$. Here, K' is the number of reconstructions used as samples after pooling. For DDRM, $K' = K$ since each run produces one reconstruction. For FPS SMC, $K' = KM$ when pooling M particles across K runs.

Let (h, w, c) index pixels and channels, and flatten to $j \in \{1, \dots, P\}$. Let $x_j^{(k)}$ be the value of sample k at index j. The sample mean is

$$\hat{\mu}_j = \frac{1}{K'} \sum_{k=1}^{K'} x_j^{(k)}, \tag{14}$$

and the unbiased sample variance is

$$\hat{v}_j = \frac{1}{K' - 1} \sum_{k=1}^{K'} \left(x_j^{(k)} - \hat{\mu}_j \right)^2. \tag{15}$$

The mean image is $\hat{\mu}$ is used as a point estimate for accuracy metrics. The variance image $\hat{v}$ is used as a basic uncertainty map.

3.4 Point estimate accuracy

Although this study focuses on uncertainty, we also report point estimate accuracy on the sample mean. Let $\hat{\mu}$ be the sample mean reconstruction in Eq. (14) and let x^* be the ground truth image. All accuracy results are computed on the luminance Y channel. Since evaluation uses the luminance Y channel, we treat each image as single channel. Therefore $P = HW$ in the accuracy metrics below.

Mean squared error. The mean squared error is

$$\mathrm{MSE}(\hat{\mu}, x^*) = \frac{1}{P} \sum_{j=1}^{P} \left(\hat{\mu}_j - x_j^* \right)^2, \tag{16}$$

where $P = HW$ is the number of pixels.

PSNR. Peak signal-to-noise ratio (PSNR) is commonly used in image restoration to report fidelity in dB [24]. It is defined as

$$\mathrm{PSNR}(\hat{\mu}, x^*) = 10 \log_{10} \left(\frac{L^2}{\mathrm{MSE}(\hat{\mu}, x^*)} \right), \tag{17}$$

where L is the data range. Since images are clipped to $[0, 1]$, we use $L = 1$.

SSIM. Structural similarity (SSIM) compares local image statistics and is widely used as a perceptual quality metric [25]. For two local patches with means $\mu_{\hat{\mu}}$ and μ_{x^*}, variances $\sigma_{\hat{\mu}}^2$ and $\sigma_{x^*}^2$, and covariance $\sigma_{\hat{\mu}x^*}$, SSIM is

$$\mathrm{SSIM} = \frac{(2\mu_{\hat{\mu}}\mu_{x^*} + C_1)(2\sigma_{\hat{\mu}x^*} + C_2)}{(\mu_{\hat{\mu}}^2 + \mu_{x^*}^2 + C_1)(\sigma_{\hat{\mu}}^2 + \sigma_{x^*}^2 + C_2)}. \tag{18}$$

We use the standard constants $C_1 = (k_1 L)^2$ and $C_2 = (k_2 L)^2$ with $k_1 = 0.01$, $k_2 = 0.03$, and $L = 1$ [25].

3.5 Calibration and reliability curves

Calibration describes whether theoretical probabilistic statements match empirical evaluations. A common notion for continuous outputs is interval calibration. If a method outputs a nominal credible interval level, then the ground truth should fall inside that interval with a similar empirical proportion. This idea is standard in calibrated regression and probabilistic forecasting [17, 26]

In this project, we represent uncertainty as variability across solver outputs for a fixed measurement. We evaluate marginal calibration using a reliability/calibration curve. This kind of curve plots empirical coverage against the nominal interval level $(1 - \alpha)$. A well-calibrated method lies close to the diagonal. The UQ benchmarking study suggests coverage-based evaluation because point estimate accuracy, like PSNR, alone does not validate uncertainty behaviour for PnPDP [14].

This evaluation is marginal since we check each pixel value separately. It does not test joint spatial calibration across pixels. We keep this focus because marginal calibration is already informative and it is widely used in regression uncertainty evaluations, while the joint spatial analysis is left for future work [14, 17].

3.5.1 Pixelwise credible intervals

Let $\gamma \in (0,1)$ denote a nominal coverage level, for example $\gamma = 0.95$. For each index j, an interval $I_j(\gamma)$ is constructed from the pooled sample set $\{x_j^{(k)}\}_{k=1}^{K'}$.

Quantile interval. Let $\widehat{Q}_j(q)$ be the empirical q quantile of $\{x_j^{(k)}\}_{k=1}^{K'}$. The two sided quantile interval is

$$I_j^{\mathrm{q}}(\gamma) = \left[\widehat{Q}_j\left(\frac{1-\gamma}{2}\right), \widehat{Q}_j\left(\frac{1+\gamma}{2}\right)\right]. \tag{19}$$

Gaussian interval. A second construction uses Gaussian approximation with mean $\hat{\mu}_j$ and variance $\hat{v}_j$. Let Φ be the standard normal cumulative distribution function and let

$$z_\gamma = \Phi^{-1}\left(\frac{1+\gamma}{2}\right). \tag{20}$$

The Gaussian interval is

$$I_j^{\mathrm{g}}(\gamma) = \left[\hat{\mu}_j - z_\gamma\sqrt{\hat{v}_j}, \hat{\mu}_j + z_\gamma\sqrt{\hat{v}_j}\right]. \tag{21}$$

3.5.2 Coverage and reliability curve

Let x^* be the ground truth HR image and let x_j^* be its pixel value at index j. Let $\mathbb{I}(\cdot)$ be the indicator function. For a single image, empirical coverage at nominal level γ is

$$\widehat{\mathrm{Cov}}(\gamma) = \frac{1}{P}\sum_{j=1}^{P}\mathbb{I}\big(x_j^* \in I_j(\gamma)\big). \tag{22}$$

A reliability curve plots $\widehat{\mathrm{Cov}}(\gamma)$ against γ over a set of nominal levels. A calibrated method should lie close to the diagonal line $\widehat{\mathrm{Cov}}(\gamma) = \gamma$ [14, 17].

3.5.3 Calibration error

We also report a metric that is a calibration error, which averages the deviation from the diagonal line.

Let Γ be a set of nominal levels, for example $\Gamma = \{0.05, 0.10, \ldots, 0.95\}$. Calibration error is

$$\mathrm{CalErr} = \frac{1}{|\Gamma|}\sum_{\gamma\in\Gamma}\left|\widehat{\mathrm{Cov}}(\gamma) - \gamma\right|. \tag{23}$$

Smaller values indicate better marginal calibration. [17]

3.6 Sharpness and interval width

Calibration alone is not sufficient, since, trivially, a method can achieve high coverage simply by outputting very wide intervals. Therefore, probabilistic forecast evaluation often follows the principle of maximizing sharpness subject to calibration [26]. Sharpness describes how concentrated the predicted distribution is. In practice, we use interval width as a simple sharpness approximation at a fixed nominal level [17].

For an interval $I_j(\gamma) = [\ell_j(\gamma), u_j(\gamma)]$, define pixelwise width

$$w_j(\gamma) = u_j(\gamma) - \ell_j(\gamma). \tag{24}$$

So, for an image, the mean width is

$$W(\gamma) = \frac{1}{P}\sum_{j=1}^{P} w_j(\gamma). \tag{25}$$

Interval width complements the reliability curve by clarifying whether miscalibration arises from intervals that are too narrow or too wide; the former often indicates overconfidence, whereas the latter suggests insufficient confidence.

3.7 Usefulness metrics

Calibration measures probability correctness, but it does not capture where the reconstruction is likely to be wrong. We call this second property "usefulness". A useful uncertainty map assigns larger uncertainty to regions where the solver tends to make larger errors. The UQ benchmarking study argues that reconstruction accuracy and uncertainty are not necessarily linked across solvers, so uncertainty should be evaluated explicitly and in multiple ways [14].

We evaluate usefulness using a rank-based association between predicted uncertainty and observed error magnitude. In particular, we use Spearman rank correlation between pixel-wise absolute error and pixel-wise uncertainty/variance. Spearman correlation measures whether two quantities have a monotonic relationship by computing correlation on their ranks rather than on their raw values. This type of metric is commonly used in regression uncertainty evaluation as a way to test whether uncertainty ranks difficult cases higher [27, 28, 16].

Formally, for each index j, define the absolute error of the posterior mean

$$e_j = \left|\hat{\mu}_j - x_j^*\right|, \tag{26}$$

the uncertainty is $\hat{v}_j$.

Let $\mathrm{rank}(\cdot)$ map a vector to its ranks, with ties handled by average ranks. Let $r_e = \mathrm{rank}(e)$ and $r_v = \mathrm{rank}(\hat{v})$. The Spearman correlation is the Pearson correlation of these ranks,

$$\rho_{\mathrm{sp}} = \mathrm{corr}(r_e, r_v). \tag{27}$$

A positive value means that large uncertainty is likely imply large error in a rank sense. However, it is worth mentioning that usefulness does not imply calibration. A method can rank errors well but still produce intervals that are either too narrow or too wide.

3.8 Observed and Null-ish Space Analysis for SR

SR is closely linked to frequency content. A typical degradation model applies blur and subsampling, thereby diminishing and discarding high-frequency information. This is a classical reason why SR is ill-posed [1, 2]. A sampling theorem view provides a related intuition: when an image is observed at a lower sampling, only a limited frequency band can be clearly recovered, while higher frequency content is weakly constrained and prone to aliasing, which occurs when downsampling causes high-frequency content to be misrepresented as lower frequency patterns, so different high-frequency details can produce similar LR observations [1, 29].

The UQ benchmarking study discusses a related idea using observable and null directions in linear inverse problems [14]. The key message is that uncertainty should be larger in directions that are weakly constrained by the measurement. In SR, an explicit null space analysis is less trivial for a practical bicubic operator. Therefore, a frequency domain proxy is used. Low frequencies are treated as more observed by the measurement, while high frequencies are treated as more null-ish. This section introduces a task-specific test that checks whether solver-induced uncertainty is allocated to the weakly constrained frequency region.

3.8.1 Spectra for uncertainty and error

To test this behaviour, we introduce a frequency domain based on the discrete Fourier transform. We compute Fourier transforms using the fast Fourier transform, which reduces the cost of computing discrete Fourier coefficients on a grid [30]. Let $\mathcal{F}$ be the two-dimensional discrete Fourier transform applied to each channel. Let ω index discrete spatial frequencies.

The analysis uses two spectra. The first describes how sample variability is distributed across frequencies. The second describes how the reconstruction error of the sample mean is distributed across frequencies.

Residual uncertainty spectrum. Define the residual of pooled sample k as

$$r^{(k)} = x^{(k)} - \hat{\mu}. \tag{28}$$

This residual removes the shared mean structure and keeps the variation across samples. The residual uncertainty spectrum is the average residual power spectrum, i.e.

$$S(\omega) = \frac{1}{K'} \sum_{k=1}^{K'} \left|\mathcal{F}(r^{(k)})(\omega)\right|^2. \tag{29}$$

Large values of $S(\omega)$ at high frequencies indicate that samples vary mainly in fine details. Large values at low frequencies indicate that samples vary mainly in coarse structure.

Error spectrum. Define the mean error image as

$$\varepsilon = \hat{\mu} - x^*. \tag{30}$$

The error spectrum is

$$E(\omega) = \left|\mathcal{F}(\varepsilon)(\omega)\right|^2. \tag{31}$$

Large values of $E(\omega)$ at high frequencies indicate that the mean reconstruction is wrong mainly in fine details. Large values at low frequencies indicate that the mean reconstruction is wrong mainly in coarse structure.

3.8.2 Energy normalisation and summary ratios

To compare spectral shapes rather than overall scale, spectra are normalised by total energy. Let Ω be the set of discrete frequencies. Define

$$\bar{S}(\omega) = \frac{S(\omega)}{\sum_{\omega' \in \Omega} S(\omega')}, \qquad \bar{E}(\omega) = \frac{E(\omega)}{\sum_{\omega' \in \Omega} E(\omega')}. \tag{32}$$

Observed and null-ish frequency bands. Let s denote the super-resolution scale factor. The LR grid has Nyquist frequency 0.5 cycles per LR pixel. Expressed in HR pixel units, this gives the cutoff

$$f_c = \frac{0.5}{s}, \tag{33}$$

which we use as a proxy boundary between more observed and more null-ish frequencies [31].

High to low ratios. Define the low and high frequency sets by

$$\Omega_{\mathrm{L}} = \{\omega \in \Omega : |\omega| < f_c\}, \qquad \Omega_{\mathrm{H}} = \Omega \setminus \Omega_{\mathrm{L}}. \tag{34}$$

Then define the high-to-low ratios for uncertainty and error as

$$R_{\mathrm{U}} = \frac{\sum_{\omega \in \Omega_{\mathrm{H}}} \bar{S}(\omega)}{\sum_{\omega \in \Omega_{\mathrm{L}}} \bar{S}(\omega)}, \qquad R_{\mathrm{E}} = \frac{\sum_{\omega \in \Omega_{\mathrm{H}}} \bar{E}(\omega)}{\sum_{\omega \in \Omega_{\mathrm{L}}} \bar{E}(\omega)}. \tag{35}$$

Under exploration ratio. The key diagnostic compares these two ratios,

$$R_{\mathrm{UE}} = \frac{R_{\mathrm{U}}}{R_{\mathrm{E}}}. \tag{36}$$

This ratio is introduced in this paper as a task specific diagnostic for super resolution, motivated by the observable versus null direction analysis in [14]. If $R_{\mathrm{UE}} < 1$, then uncertainty allocates relatively less high-frequency energy than the reconstruction error. This indicates that the sampler explores high-frequency ambiguity less than the error pattern suggests. In super resolution, this can appear as visually plausible but incorrect fine detail without a corresponding uncertainty warning. If $R_{\mathrm{UE}} \approx 1$, then uncertainty and error have similar high-frequency emphasis. If $R_{\mathrm{UE}} > 1$, then uncertainty is more high-frequency than the error, which indicates conservative uncertainty allocation in fine details.

4 Methods

4.1 Problem Setting and Notation

This section fixes the notation used in the rest of this section. Section 2 gives the background on SR as an inverse problem and on the Bayesian framework. Here, we only restate what is needed to define the methods.

4.1.1 Images and operator

Let $x \in [0,1]^{H \times W \times C}$ denote an HR image with height H, width W, and C channels. Let $y \in [0,1]^{h \times w \times C}$ denote the observed LR measurement. Let $s \in \{4, 8\}$ denote the scale factor, so $h = H/s$ and $w = W/s$.

Let A_s represent bicubic downsampling at scale s. It maps an HR image to an LR image,

$$A_s : \ [0,1]^{H \times W \times C} \to [0,1]^{h \times w \times C}.$$

4.1.2 Degradation model and noise

Let x^* be the ground truth HR image. We use the model as

$$y = A_s x^* + \epsilon, \tag{37}$$

where ϵ is measurement noise.

When $\sigma_0 > 0$, we use additive Gaussian noise,

$$\epsilon \sim \mathcal{N}(0, \sigma_0^2 I), \tag{38}$$

where σ_0 is the noise standard deviation and I is the identity matrix of matching dimension. This gives the likelihood as

$$p(y \mid x) \propto \exp\left(-\frac{1}{2\sigma_0^2} \|y - A_s x\|_2^2\right). \tag{39}$$

When $\sigma_0 = 0$, then the measurement is noise-free. In this case, the data term is a hard constraint, which means the reconstruction is required to satisfy

$$y = A_s x \tag{40}$$

exactly. Equivalently, a set of feasible images is restricted to those that are consistent with the measurement under operator A_s. The solvers can still be stochastic because they use a generative prior, but they should not violate (40).

4.1.3 Prior, posterior and samples

Let $p_\theta(x)$ be an implicit natural image prior given by a pretrained diffusion model. The Bayesian posterior is

$$p_\theta(x \mid y) \propto p(y \mid x)\, p_\theta(x). \tag{41}$$

Each inverse solver defines a conditional distribution over reconstructions for a fixed measurement y. To approximate this distribution, we run the solver multiple times with different random seeds. Let $x^{(k)}$ denote the reconstruction from run k, for $k \in \{1, ..., K\}$. The sample set is defined as

$$\mathcal{S}(y) = \{x^{(1)}, x^{(2)}, \ldots, x^{(K)}\}. \tag{42}$$

When $\sigma_0 > 0$, the measurement y is generated once per image using a fixed measurement seed. Then, this same y is reused across all runs. This makes repeated runs correspond to a single conditional input, so variation across $\{x^{(k)}\}$ reflects solver randomness/uncertainty rather than changes in the measurement. The next sections define how $\mathcal{S}(y)$ is summarised and how uncertainty is evaluated.

We use K to denote the number of solver runs. When a solver produces multiple particles per run, we pool reconstructions and denote the pooled sample count by K' (defined in Section 3).

4.2 Sampling and evaluation pipeline

This section summarises how reconstructions are generated and evaluated in our experiments. For each image $x^{*(n)}$ and setting (s, σ_0), we form a single measurement $y^{(n)}$ using the bicubic operator A_s. When $\sigma_0 > 0$, we fix the measurement seed so that the same $y^{(n)}$ is reused across all stochastic runs for that image. When $\sigma_0 = 0$, $y^{(n)}$ is deterministic given $x^{*(n)}$ and A_s. We then vary only the solver randomness (run seed, and particles for FPS-SMC) to obtain a pooled set of reconstructions.

For DDRM, each run produces one reconstruction, so the pooled sample set is $\mathcal{S}^{(n)}_{\mathrm{DDRM}} = \{x^{(n,k)}\}_{k=1}^{K}$. For FPS-SMC, each run produces M particles, so $\mathcal{S}^{(n)}_{\mathrm{FPS}} = \{x^{(n,k,m)}\}_{k=1,\ldots,K;\, m=1,\ldots,M}$. From each pooled sample set, we compute the sample mean $\hat{\mu}^{(n)}$ and sample variance $\hat{v}^{(n)}$. These outputs are then scored using the accuracy and uncertainty metrics defined in Section 3. Algorithm 1 provides the full procedure.

The remainder of this section describes the two solvers, DDRM and FPS-SMC, and explains how each produces samples for the pipeline above.

Algorithm 1 Sampling and evaluation pipeline for SR uncertainty experiments

Inputs: Dataset $\mathcal{D} = \{x^{*(n)}\}_{n=1}^{N}$; settings $\mathcal{G} = \{(s, \sigma_0)\}$; run counts K_{DDRM} and K_{FPS}; reverse steps T (respaced); particle count M (FPS-SMC only); nominal levels Γ
Outputs: Dataset level metrics aggregated over images

```
for all (s, σ_0) ∈ G do
    for n = 1 to N do
        Draw ε^(n) ~ N(0, I) and generate measurement once:
            y^(n) ← A_s x^*(n) + σ_0 ε^(n)
        Initialise S^(n)_DDRM ← ∅
        Initialise S^(n)_FPS ← ∅
        for k = 1 to K_DDRM do
            Run DDRM with T reverse steps to obtain one sample x^(n,k)
            S^(n)_DDRM ← S^(n)_DDRM ∪ {x^(n,k)}
        end for
        if σ_0 > 0 then
            for k = 1 to K_FPS do
                Run FPS-SMC with T reverse steps to obtain particles {x^(n,k,m)}_{m=1}^M
                S^(n)_FPS ← S^(n)_FPS ∪ {x^(n,k,m)}_{m=1}^M
            end for
        end if
        Compute sample mean μ̂^(n) and sample variance v̂^(n) from each sample set
        Compute per image metrics using Section 3, including: PSNR and SSIM of μ̂^(n), coverages {Cov^(n)(γ)}_{γ∈Γ},
        interval widths, Spearman correlation, and frequency ratios
    end for
    Average per image metrics over n = 1, ..., N to obtain dataset level results
end for
```

4.3 DDRM for Super Resolution

This section describes Denoising Diffusion Restoration Models (DDRM) for SR. Section 2 introduced diffusion models as priors and gave a taxonomy of inverse solvers and uncertainty. Here, DDRM is presented as a detailed algorithm and as the main solver in our experiments. The description follows the DDRM paper [9].

4.3.1 Overview of the sampling process

DDRM aims to generate HR reconstructions that are realistic under a diffusion prior and consistent with the LR measurement. At inference time, DDRM runs a reverse diffusion chain. For each reverse time step t, the pretrained diffusion model produces a prior prediction based on the current latent state x_t. DDRM then applies a data-consistency update that injects the fixed measurement y. This is the main structural difference from unconditional DDPM sampling in Fig. 2, where no measurement is used. After the data-consistency update, the next latent x_{t-1} is sampled using a diffusion transition that adds back random Gaussian noise to keep the diffusion noise schedule and keep the process stochastic. Figure 4 summarises this structure.

4.3.2 Conditional diffusion view and variational objective

DDRM is motivated by a conditional diffusion model for the posterior distribution. Let y be a fixed measurement from the linear model in Section 4.1. DDRM defines a conditional Markov chain

$$p_\theta(x_{0:T} \mid y) = p(x_T) \prod_{t=1}^{T} p_\theta(x_{t-1} \mid x_t, y), \tag{43}$$

where x_T is an initial noise sample and x_0 is the final reconstruction. The aim is that the marginal $p_\theta(x_0 \mid y)$ approximates the Bayesian posterior.

To train such a conditional diffusion model, the DDRM paper introduces a factorised variational distribution

$$q(x_{1:T} \mid x_0, y) = \prod_{t=1}^{T} q(x_t \mid x_{t-1}, x_0, y), \tag{44}$$

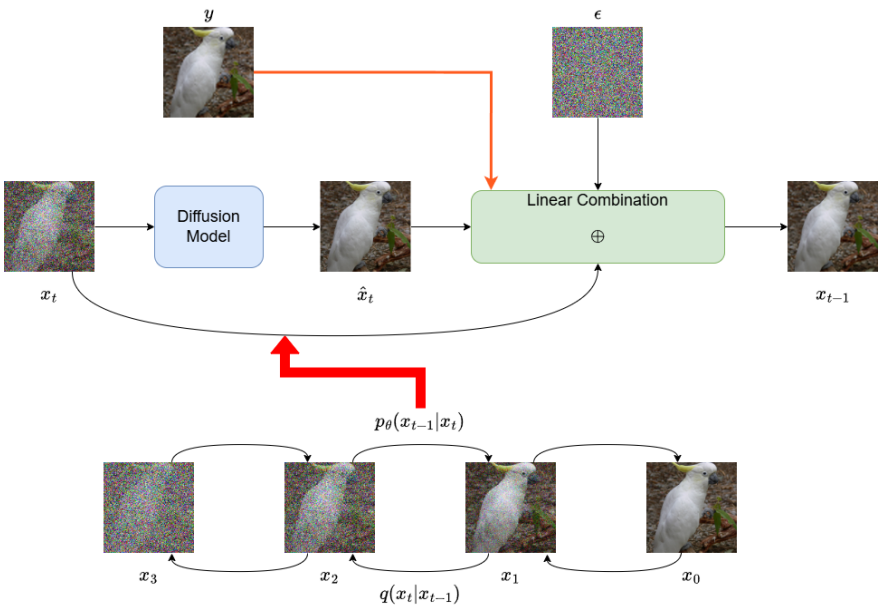


Figure 4: DDRM sampling at one reverse step. The diffusion model produces a prior prediction from the current latent x_t. DDRM then injects the fixed measurement y into a data consistency update (orange arrow) before sampling x_{t-1}. This measurement injection is the key difference from unconditional DDPM sampling in Figure 2. Figure created by the author, inspired by [9].

which produces an ELBO for $\log p_\theta(x_0 \mid y)$ [9]. The key design is to construct $q(\cdot)$ and $p_\theta(\cdot)$ so that a pretrained unconditional diffusion model remains a good solution to the conditional objective.

This variational viewpoint is important for this paper. It clarifies that DDRM defines a solver-induced conditional distribution. Therefore, calibration is not assumed. Instead, calibration is evaluated empirically in Section 6 using the framework in Section 3.

4.3.3 SVD spectral space

DDRM assumes a known linear degradation operator. In our experiments, the operator is bicubic downsampling at scale s, denoted by A_s. DDRM uses a singular value decomposition (SVD) of the operator,

$$A_s = U\Sigma V^\top, \tag{45}$$

where U and V are orthogonal and Σ is diagonal and rectangular. Let s_i denote the i-th singular value of A_s.

Define spectral coordinates

$$\bar{x} = V^\top x, \qquad \bar{y} = U^\top y, \qquad \bar{\epsilon} = U^\top \epsilon. \tag{46}$$

Using the measurement model $y = A_s x^* + \epsilon$, the i-th spectral coefficient then becomes

$$\bar{y}_i = s_i \bar{x}_i^* + \bar{\epsilon}_i. \tag{47}$$

This diagonal form means that each measurement $\bar{y}$ coefficient depends only on one coefficient of the unknown image $\bar{x}$, which supports DDRM closed-form updates.

The singular values define which directions are informed by the measurement. Let

$$\mathcal{I}_{\text{obs}} = \{i : s_i > 0\}, \qquad \mathcal{I}_{\text{null}} = \{i : s_i = 0\}. \tag{48}$$

For $i \in \mathcal{I}_{\text{null}}$, the measurement provides no information about $\bar{x}_i$. For $i \in \mathcal{I}_{\text{obs}}$, define the normalized observation

$$\tilde{y}_i = \frac{\bar{y}_i}{s_i}. \tag{49}$$

When the measurement noise standard deviation is σ_0, the effective noise level for $\tilde{y}_i$ is σ_0/s_i. Small s_i therefore corresponds to weak measurement information. This links to the null-ish viewpoint in Section 3.8 for SR, where ambiguity is expected to concentrate in weakly observed directions.

4.3.4 Gaussian marginals and the measurement update

DDRM constructs conditional transitions in spectral space. The construction is designed to preserve a property called Gaussian marginals. Informally, the aim is that for each time t, the conditional marginal $q(\bar{x}_t \mid x_0, y)$ remains Gaussian with a variance that matches the diffusion noise schedule. In the DDRM paper, this is formalised by a set of conditional Gaussian transitions for each index i and by Proposition 3.1 in the paper [9].

This construction leads to a practical update rule inside the reverse diffusion chain. At reverse step t, the pretrained diffusion model provides a prediction of the clean image from the current latent state,

$$\hat{x}_0 = \hat{x}_0(x_t). \tag{50}$$

This estimate is mapped to spectral space, $\widehat{\bar{x}}_0 = V^\top \hat{x}_0$. DDRM then aligns the estimate with the measurement in the observable directions. The alignment depends on three quantities: the singular value s_i, the measurement noise σ_0, and the diffusion noise level at step t. Denote the diffusion noise scale at step t by σ_t. DDRM splits the alignment into several cases depending on whether $s_i = 0$ and on the comparison between σ_t and σ_0/s_i. The cases can be interpreted as follows:

Unobserved directions. If $i \in \mathcal{I}_{\text{null}}$, then $s_i = 0$ and the measurement contains no information. In this case, DDRM simply keeps the prior prediction in these coefficients. These directions are purely governed by the diffusion prior.

Observed directions with strong measurement signal. If $i \in \mathcal{I}_{\text{obs}}$ and the effective measurement noise σ_0/s_i is small relative to the diffusion noise σ_t, then the latent is noisier than the measurement in that coefficient. DDRM therefore trusts the measurement more in that direction and injects information from $\tilde{y}_i$.

Observed direction with weak measurement signal. If $i \in \mathcal{I}_{\text{obs}}$ and the effective measurement noise σ_0/s_i is large relative to the diffusion noise σ_t, then the measurement is noisier than the latent in that coefficient. DDRM therefore relies more on the diffusion prior prediction and makes a weak alignment towards the measurement.

The output of this step is a corrected spectral estimate $\tilde{\bar{x}}_0$. Mapping back gives $\tilde{x}_0 = V\tilde{\bar{x}}_0$. DDRM then samples the next latent x_{t-1} using a DDPM normal reverse transition centred at $\tilde{x}_0$. This preserves the diffusion noise schedule while enforcing measurement consistency. For readability, the full update formulas are not repeated here. They can be found in Section 3.2 of the DDRM paper [9].

4.3.5 Why a pretrained DDPM can be reused

A central result in DDRM is that the conditional training objective can match the unconditional objective under assumptions, which is Theorem 3.2. in the paper. The theorem states that under conditions on the diffusion parameterisation and weight sharing across time steps, the DDRM ELBO can be rewritten in the form of the DDPM objective [9]. This is the reason why a pretrained unconditional diffusion model can be used without retraining on a specific inverse task. So, in practice, DDRM uses the pretrained model to produce $\hat{x}_0(x_t)$, and only changes the measurement update via the SVD of the degradation operator.

These points matter for uncertainty quantification, even when the measurement model is correct, the solver-induced distribution can deviate from the true posterior in practice, because it depends on the reverse chain discretisation and other design choices [9, 14]. We fix the DDRM hyperparameters to the published reference implementation and do not study their sensitivity, which is left for future work. We then evaluate uncertainty using calibration, sharpness, usefulness, and frequency tests, which are defined in Section 3

4.3.6 Taxonomy

After explaining the algorithm, DDRM can be placed in the taxonomies from Sec. 3.2.

Structural taxonomy. DDRM uses the explicit linear operator and performs a measurement correction in the SVD spectral basis. The update depends directly on the singular values of A_s, since this is the key design of DDRM. So, DDRM is categorised as a linear guidance method.

Uncertainty taxonomy. DDRM produces diverse samples for fixed y. because the reverse diffusion process includes random sampling noise. Therefore, it defines a conditional distribution over outputs, rather than a single deterministic estimate.

Section 2.3 introduced a score-based view of Bayesian inference, where the posterior score is decomposed as Eq. (12). For a diffusion-based solver, we just replace x with x_t, the latent state variable at reverse step t. The prior term corresponds to a score under the diffusion prior and can be obtained from the pretrained unconditional model. The likelihood term is more subtle in linear inverse tasks like SR. The forward model specifies the likelihood $p(y \mid x_0)$ for the clean image x_0, not for the noisy latent x_t. The induced likelihood $p(y \mid x_t)$ would require marginalizing the unknown clean image,

$$p(y \mid x_t) \;=\; \int p(y \mid x_0)\, p(x_0 \mid x_t)\, dx_0,$$

so its gradient is not available simply by applying data consistency in x_t.

DDRM avoids explicitly computing this likelihood score. Instead, under the linear Gaussian forward model and the Gaussian structure of the diffusion noising process, the pair (x_t, y) is jointly Gaussian. This results in a closed-form conditioning step in spectral space defined by the SVD decomposition of the operator. The resulting algorithm induces a conditional distribution that depends on this conditional construction and on the discretisation used for inference. Thus, DDRM is treated as a heuristic posterior approximation in the uncertainty taxonomy [14]. Therefore, calibration is treated as a measurable property rather than an assumed theoretical guarantee.

4.4 Filtering Posterior Sampling with sequential Monte Carlo

This section explains Filtering Posterior Sampling (FPS) and its sequential Monte Carlo variant (FPS-SMC) as included in our experiments. FPS is chosen as a comparison method because it is designed for posterior targeting under linear Gaussian likelihood, using a Bayesian filtering viewpoint and sequential Monte Carlo [18]. The description here follows the original FPS paper and the reference implementation.

4.4.1 Overview

FPS views diffusion sampling as a sequence of latent states $\{x_t\}_{t=0}^{T}$ evolving over reverse time. The central idea is to treat conditioning on a fixed measurement y as a filtering problem defined along this reverse time axis. This produces a sequence of intermediate conditional target distributions for the latents. FPS then uses sequential Monte Carlo to approximate these targets using a weighted particle design with resampling [18].

In this study, FPS-SMC is used as a comparison solver because its sampling mechanism is based on sequential Monte Carlo, which incorporates the measurement through importance weighting and resampling, providing an explicit correction step towards the intended conditional targets. Therefore, FPS-SMC is expected to be closer to posterior targeting than guidance-based heuristic solvers such as DDRM. However, it is still evaluated empirically. The theoretical motivation is asymptotic in the particle limit, while our experiments use a finite number of particles.

FPS-SMC relies on the Gaussian measurement model defined in Sec. 4.1. In particular, the likelihood assumes additive Gaussian noise with positive noise level $\sigma_0 > 0$. Therefore, FPS-SMC is evaluated only in the noisy settings in our experiments.

4.4.2 Filtering construction and sequential Monte Carlo algorithm

Coupled noisy measurement chain. A key ingredient of FPS is a coupled construction that introduces a sequence of noisy measurements $\{y_t\}$ aligned with the diffusion noise level. This allows measurement information to be injected gradually across the reverse process [18].

Write the diffusion forward process in generic linear Gaussian form,

$$x_t = a_t x_{t-1} + b_t z_t, \qquad z_t \sim \mathcal{N}(0, I), \tag{51}$$

where a_t and b_t are determined by the diffusion noise schedule. FPS defines a corresponding noisy measurement chain by using the same noise samples z_t passed through the forward model, so

$$y_0 = y, \qquad y_t = a_t y_{t-1} + b_t A_s z_t. \tag{52}$$

This makes the noise in y_t correlated with the latent noise in x_t. The observed input remains $y_0 = y$. The role of $\{y_t\}$ is to provide intermediate conditioning variables for the filtering updates.

Conditional transition and proposal. At reverse step t, FPS conditions the latent update on the current measurement state y_{t-1}. The conditional reverse transition used by FPS satisfies the factorisation

$$p_\theta(x_{t-1} \mid x_t, y_{t-1}) \;\propto\; p_\theta(x_{t-1} \mid x_t)\, p_\theta(y_{t-1} \mid x_{t-1}), \tag{53}$$

which follows from Bayes' rule. The term $p_\theta(x_{t-1} \mid x_t)$ is the reverse transition of the pretrained diffusion sampler. Under standard DDPM parameterisations, it is Gaussian. The term $p_\theta(y_{t-1} \mid x_{t-1})$ acts as a measurement likelihood at the same noise level. So the conditional reverse transition is implied by the linear Gaussian measurement model together with the coupled measurement construction. Therefore, it is also a tractable Gaussian under these assumptions.

In FPS-SMC, particles are generated by sampling a proposal distribution. In the FPS-SMC algorithm, the proposal is taken to be the conditional transition itself,

$$q_t(\cdot \mid x_t, y_{t-1}) \;=\; p_\theta(\cdot \mid x_t, y_{t-1}). \tag{54}$$

The word *proposal* refers to the distribution used to draw candidate particles. After propagation, importance weights and resampling correct the particle system so that it approximates the intended filtering targets [18].

SMC particle updates. Sequential Monte Carlo represents each intermediate conditional distribution using M particles with weights. At time t, the particle set is $\{x_t^{(m)}\}_{m=1}^M$ with non negative weights $\{w_t^{(m)}\}_{m=1}^M$. Normalised weights are

$$\tilde{w}_t^{(m)} = \frac{w_t^{(m)}}{\sum_{j=1}^M w_t^{(j)}}. \tag{55}$$

Each reverse step has three operations as follows:

Propagation:

$$x_{t-1}^{(m)} \sim q_t(\cdot \mid x_t^{(m)}, y_{t-1}), \qquad m = 1, \dots, M. \tag{56}$$

Resampling weights: weights are updated using the resampling probability rule derived in the FPS paper. For readability, the full expression is not repeated here. It is given in [18] and implemented in the reference code.

Resampling: resampling draws indices according to the normalised weights to reduce weight degeneracy, which means that after several steps, most importance weight is carried only by a small number of particles. Equivalently, many particles have negligible weight and contribute little to estimates. This reduces the effective diversity of the particle set and can make the approximation unstable.

This is usually addressed using the effective sample size,

$$\mathrm{ESS}_{t-1} = \frac{1}{\sum_{m=1}^M \left(\tilde{w}_{t-1}^{(m)}\right)^2}. \tag{57}$$

If all weights are equal, then $\mathrm{ESS}_{t-1} = M$. If one particle dominates the weights, then ESS_{t-1} approaches 1. When ESS_{t-1} is small, resampling duplicates high weight particles and discards low weight particles. This reduces degeneracy but introduces additional Monte Carlo variability.

4.4.3 Taxonomy

Structural taxonomy. FPS-SMC is categorised as a sequential Monte Carlo based diffusion solver. It propagates particles through reverse diffusion time and corrects them using importance weights and resampling. It requires applying the forward operator A_s and evaluating Gaussian likelihood terms implied by the degradation model in Section 4.1. Unlike DDRM, it does not rely on an SVD-based diagonalisation of the operator.

Uncertainty taxonomy. FPS-SMC is treated as posterior targeting in an asymptotic sense, since SMC methods can converge to the target distribution as the particle count increases under stated assumptions [18]. In our experiments, the particle count is finite. Therefore, approximation error can arise due to weight degeneracy, resampling variability, and the discretisation used for the reverse diffusion chain. For this reason, calibration and usefulness are evaluated empirically rather than assumed a priori.

5 Experimental Setup

This section specifies the concrete experimental choices used to evaluate uncertainty for SR. Section 4 defines the forward model, notation, and solvers. Section 3 defines all evaluation metrics. Here we only describe the dataset, the settings, and the sampling protocol.

5.1 Dataset and Preprocessing

Experiments use ImageNet images at resolution 256×256. Images are normalised to the range $[0, 1]$. We evaluate on $N = 200$ images, denoted $\{x^{*(n)}\}_{n=1}^{N}$.

The low resolution measurement $y^{(n)}$ is generated from $x^{*(n)}$ using the forward model in Eq. (37) with operator A_s and noise model in Eq. (38). The measurement is then passed to the solver to generate a sample set as in Eq. (42).

For PSNR and SSIM evaluation, reconstructions are clipped to $[0, 1]$ to match the data range.

5.2 Setting Grid

We evaluate four settings defined by the scale factor s and measurement noise level σ_0:

- $s = 4$, $\sigma_0 = 0$
- $s = 4$, $\sigma_0 = 0.05$
- $s = 8$, $\sigma_0 = 0$
- $s = 8$, $\sigma_0 = 0.05$

All settings use the same $N = 200$ images. This makes comparisons across settings aligned at the image level.

5.3 Solvers Evaluated

Two diffusion-based solvers are evaluated.

DDRM. DDRM is the main solver. It uses a pretrained unconditional diffusion prior trained on ImageNet. For each image and setting, DDRM is run repeatedly with different random seeds to produce a sample set $\mathcal{S}(y^{(n)})$ as in Eq. (42).

FPS-SMC. FPS-SMC is used as a comparison solver in noisy settings. It relies on the Gaussian likelihood in Eq. (39), so it is evaluated only when $\sigma_0 > 0$. We test two particle budgets, $M \in \{8, 20\}$. To control the pooled sample count, we fix $K' = 800$ per image and choose K so that $K' = KM$.

5.4 Sampling Protocol

The key requirement for calibration metrics is that repeated runs correspond to the same conditional input. Therefore, when $\sigma_0 > 0$, the measurement $y^{(n)}$ is generated once per image using a fixed measurement seed. This same measurement is reused across all stochastic runs for that image. When $\sigma_0 = 0$, the measurement is deterministic given $x^{*(n)}$ and A_s, so this issue does not arise.

For each image n and each setting, we run each solver for K stochastic runs with different run seeds. We choose K based on an empirical variance convergence check shown in Appendix A.1, performed in the hardest setting ($s = 8$, $\sigma_0 = 0.05$). We denote by K' the number of reconstructions used as samples after pooling.

For DDRM, each run produces one reconstruction, so $K' = K$. In our main experiments, we use $K = 150$, hence $K' = 150$. For FPS-SMC, each run produces M final particles, so pooling across runs gives $K' = KM$ reconstructions. In our main experiments we fix the pooled size at $K' = 800$ and vary (K, M) as follows: $M = 8$ uses $K = 100$, and $M = 20$ uses $K = 40$. It is worth noting that particles within one SMC run are not independent due to resampling, so pooling increases K' but does not necessarily increase the effective number of independent samples by the same factor.

5.5 Compute Budget

Both solvers generate samples by running a reverse diffusion chain. We control inference cost by fixing the number of reverse steps. Let T denote the number of reverse steps used at inference time. All main experiments use $T = 20$ reverse steps with a fixed respacing schedule.

This choice is a compute control decision. In a small step sensitivity check, increasing T beyond 20 changed the PSNR of the sample mean only slightly. Therefore, $T = 20$ is used for the main grid to support repeated stochastic runs at fixed cost. Since the solver-induced distribution can depend on discretisation, the use of $T = 20$ is treated as a limitation in later sections.

5.6 Aggregation and Metric Computation

The sampling and pooling steps follow Algorithm 1 in Section 4.2. For each image n, the sample set $\mathcal{S}(y^{(n)})$ is aggregated into a sample mean and an unbiased sample variance, computed elementwise, as defined in Eqs. (14) and (15). Accuracy metrics are computed from the sample mean. Calibration, sharpness, and usefulness metrics are computed from the sample set using the definitions in Section 3.

Frequency diagnostics use the residual uncertainty spectrum $S(\omega)$ in Eq. (29) and the error spectrum $E(\omega)$ in Section 3.8.1. High and low-frequency bands, and summary ratios, following Section 3.8.2. All reported results are averaged over the $N = 200$ images within each setting.

5.7 Procedure Summary

The full sampling and evaluation procedure is given in Algorithm 1 (Section 4.2). Appendix A provides additional pseudocode for coverage and frequency ratio computations.

6 Results & Discussion

This section reports the empirical behaviour of DDRM and FPS-SMC under the SR settings defined in Section 5. We focus on uncertainty behaviour, following the UQ benchmarking framework in Section 3. Results are averaged over $N = 200$ test images from ImageNet. When we summarise variation across images, and report the standard deviation across the N image level metrics.

For each image and solver, we generate a sample set of reconstructions conditioned on a fixed measurement y. The sample mean $\hat{\mu}$ is used as a point estimate for accuracy. Uncertainty is represented by the unbiased pixelwise sample variance $\hat{v}$ and by the interval constructions in Section 3. All reported metrics are computed on the luminance Y channel unless stated otherwise.

Section 3.1 and Fig. 3 provide qualitative motivation for evaluating uncertainty. They show that similar mean reconstructions can correspond to different uncertainty maps. Thus, the remainder of this section focuses on quantitative evaluation. Additional qualitative examples are provided in Appendix B.

6.1 Accuracy on the Sample Mean

Table 1 reports PSNR and SSIM of the sample mean reconstruction $\hat{\mu}$. Values are summarised over the $N = 200$ images using the mean and standard deviation. We use K' pooled reconstructions per image, as defined in Section 5. For DDRM, $K' = 150$. For FPS-SMC, $K' = 800$.

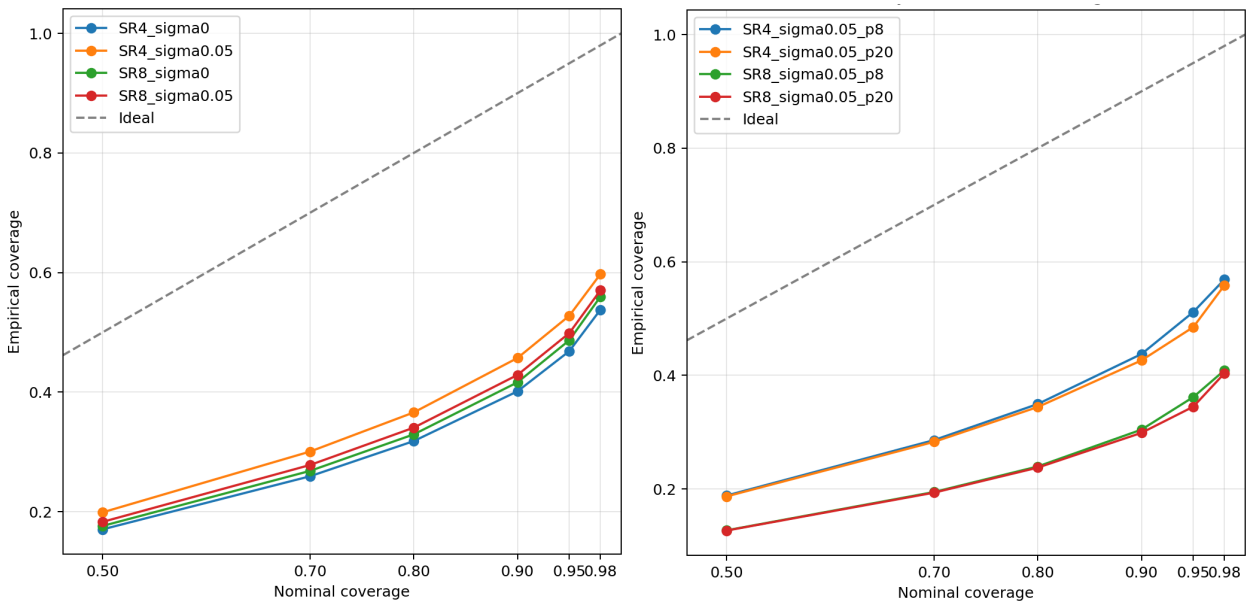
For DDRM, PSNR and SSIM decrease when the scale factor increases from $s = 4$ to $s = 8$. Increasing measurement noise also reduces the accuracy metrics. For the noisy settings, DDRM achieves higher PSNR and SSIM than FPS-SMC under the same degradation. This comparison is made under a fixed compute budget, since both solvers use $T = 20$ reverse steps in our main experiments. As discussed in Section 3, point estimate accuracy does not determine calibration or usefulness of uncertainty. We therefore evaluate uncertainty directly in the next subsections.

Table 1: Accuracy of the sample mean reconstruction $\hat{\mu}$ on the luminance Y channel. Results are reported as mean $\pm$ standard deviation over $N = 200$ images. DDRM uses $K' = 150$ pooled reconstructions per image. FPS-SMC uses $K' = 800$ pooled reconstructions per image, aggregated across particles and runs.

Method	Setting	PSNR (dB)	SSIM
DDRM	$s = 4,\ \sigma_0 = 0$	27.57 ± 4.17	0.960 ± 0.036
DDRM	$s = 4,\ \sigma_0 = 0.05$	26.26 ± 3.52	0.949 ± 0.043
DDRM	$s = 8,\ \sigma_0 = 0$	24.35 ± 3.74	0.921 ± 0.062
DDRM	$s = 8,\ \sigma_0 = 0.05$	23.49 ± 3.28	0.906 ± 0.069
FPS-SMC	$s = 4,\ \sigma_0 = 0.05$	25.31 ± 2.96	0.938 ± 0.048
FPS-SMC	$s = 8,\ \sigma_0 = 0.05$	22.66 ± 2.81	0.885 ± 0.077

6.2 Calibration and Sharpness

Here, we evaluate marginal calibration and sharpness, using the definitions in Section 3. Calibration is summarised by the reliability curve and by two scalar metrics at the nominal level $\gamma = 0.95$. These are the empirical marginal coverage $\widehat{\mathrm{Cov}}(0.95)$ and the calibration error CalErr. Sharpness is summarised by the mean interval width at $\gamma = 0.95$, reported for both quantile and Gaussian interval constructions.



(a) DDRM (all settings)

(b) FPS-SMC ($\sigma_0 = 0.05$)

Figure 5: Reliability curves for super resolution uncertainty using quantile intervals. Each curve plots empirical marginal coverage against the nominal level γ. The diagonal line indicates perfect marginal calibration. (a) DDRM results for all four settings. (b) FPS-SMC results for the noisy settings, shown for $M = 8$ and $M = 20$ particles.

Table 2: Calibration and sharpness at nominal level $\gamma = 0.95$. Coverage $\widehat{\mathrm{Cov}}(0.95)$ and CalErr use quantile intervals. Widths are reported for both quantile and Gaussian interval constructions. Results are reported as mean $\pm$ standard deviation over $N = 200$ images.

Method	Setting	$\widehat{\mathrm{Cov}}(0.95)$	CalErr	$W^{\mathrm{q}}(0.95)$	$W^{\mathrm{g}}(0.95)$
DDRM	$s = 4,\ \sigma_0 = 0.0$	0.469 ± 0.080	0.446 ± 0.005	0.042 ± 0.020	0.030 ± 0.014
DDRM	$s = 4,\ \sigma_0 = 0.05$	0.528 ± 0.085	0.397 ± 0.005	0.059 ± 0.022	0.042 ± 0.016
DDRM	$s = 8,\ \sigma_0 = 0.0$	0.487 ± 0.091	0.432 ± 0.005	0.064 ± 0.026	0.044 ± 0.018
DDRM	$s = 8,\ \sigma_0 = 0.05$	0.499 ± 0.091	0.422 ± 0.005	0.075 ± 0.027	0.054 ± 0.019
FPS-SMC ($M = 8$)	$s = 4,\ \sigma_0 = 0.05$	0.512 ± 0.122	0.415 ± 0.007	0.072 ± 0.020	0.072 ± 0.020
FPS-SMC ($M = 20$)	$s = 4,\ \sigma_0 = 0.05$	0.485 ± 0.118	0.424 ± 0.007	0.068 ± 0.018	0.071 ± 0.019
FPS-SMC ($M = 8$)	$s = 8,\ \sigma_0 = 0.05$	0.362 ± 0.107	0.532 ± 0.006	0.069 ± 0.020	0.068 ± 0.019
FPS-SMC ($M = 20$)	$s = 8,\ \sigma_0 = 0.05$	0.345 ± 0.104	0.537 ± 0.006	0.065 ± 0.018	0.068 ± 0.019

Figure 5 shows the reliability curves. Across settings, the curves lie below the diagonal, which indicates under coverage. Table 2 reports the corresponding scalar summaries.

For DDRM, coverage at $\gamma = 0.95$ ranges from 0.469 to 0.528. For $s = 4$, adding measurement noise shifts the reliability curve upwards and increases coverage. This is consistent with DDRM producing wider intervals in the noisy setting. However, coverage remains far below nominal, which indicates that the intervals are still too narrow. For $s = 8$, coverage remains close to 0.5 in both noise settings, despite wider intervals under noise. This suggests that increased uncertainty under noise does not fully compensate for the increased reconstruction difficulty at larger scale.

For FPS-SMC, reliability curves are reported only for the noisy settings and are shown for two particle budgets. At $s = 4$, coverage at $\gamma = 0.95$ lies between 0.485 and 0.512, which is similar to DDRM. At $s = 8$, coverage drops to between 0.345 and 0.362, and the reliability curve shifts downwards. The corresponding CalErr values increase at $s = 8$, which is consistent with stronger under coverage. In contrast to DDRM, the mean interval widths for FPS-SMC change only slightly between $s = 4$ and $s = 8$, and can decrease. This implies that the coverage drop is not explained by wider or narrower intervals alone. Instead, it indicates that reconstruction errors increase relative to the predicted uncertainty.

Sharpness results are reported in Table 2. For DDRM, interval widths increase with scale factor and with measurement noise, so this suggests that the solver produces broader predictive intervals in harder settings. However, coverage remains far below nominal, so the increase in width is not sufficient for calibration. Across all settings, quantile widths are larger than Gaussian widths. This indicates that pixelwise marginal sample distributions are not well approximated by a Gaussian with mean $\hat{\mu}_j$ and variance $\hat{v}_j$.

FPS-SMC is different in that quantile widths at $\gamma = 0.95$ are close to the Gaussian widths under both particle numbers. So, this is consistent with pixelwise marginals being approximately Gaussian at this nominal level. However, the mean widths remain similar when moving from $s = 4$ to $s = 8$, while coverage significantly decreases at $s = 8$. Together with the reliability curves, this indicates that at $s = 8$ the reconstruction errors increase without a corresponding increase in predicted uncertainty, leading to stronger under-coverage.

Interpretation. Qiu et al. [14] report that stochastic diffusion inverse solvers, including DDRM, can achieve coverage close to nominal in controlled diagnostics. In those diagnostics, the forward operator can be fully observed, so posterior uncertainty is dominated by measurement noise rather than by a large null space. Bicubic SR differs in that it is strongly ill-posed. Even when $\sigma_0 = 0$, many high-frequency details are not identified by the measurement. So, uncertainty depends not only on measurement noise but also on ambiguity introduced by the downsampling operator. Qiu et al. [14] emphasise that calibration can degrade when the inverse problem is more ambiguous and when uncertainty is driven by a large null space rather than by noise alone.

Our results are consistent with this perspective. The large under-coverage for DDRM suggests that the induced distribution is too concentrated for these SR settings. One plausible contributor is that the solver must represent uncertainty over many plausible high-frequency reconstructions. This is harder than representing noise driven variation when the operator is fully observed. A second contributor is the practical sampling budget. We use $T = 20$ reverse steps to control computation, which can alter the effective sampling distribution even when point metrics change only slightly. A third contributor is the solver design. DDRM combines an analytic data consistency update based on the assumed likelihood with a learned diffusion denoiser, as described in Section 4.3. In a strongly ill-posed setting, this combination may yield an approximation that underrepresents variability in components that are weakly constrained by the measurement. These points do not identify a single cause, but they motivate why calibration in SR can be substantially worse than in simpler cases.

For FPS-SMC, the sharp drop in coverage at $s = 8$ combined with nearly unchanged interval widths suggests that uncertainty does not scale with the increased error. A plausible explanation is particle degeneracy under a challenging posterior, which is discussed in the sequential Monte Carlo setting and is also mentioned in the UQ benchmarking study [14]. At larger scale, the likelihood provides weaker constraints on high-frequency components, and the posterior can be broader and more multi-modal. Under a fixed particle budget, resampling can reduce diversity and lead to underdispersion. This provides a coherent explanation for why increasing M from 8 to 20 does not lead to a clear improvement in calibration in our experiments.

Implications for using uncertainty in practice. In these experiments, neither solver provides marginally calibrated uncertainty at the nominal 95% level. For DDRM, uncertainty increases under measurement noise, but remains too small for calibration. For FPS-SMC, calibration degrades substantially at larger scale under the tested particle budgets. These results suggest that uncertainty maps from these solvers should not be interpreted as calibrated credible intervals in this SR setting. So, in later sections, we will evaluate usefulness and frequency diagnostics to assess whether uncertainty is still informative about where errors occur, even when calibration fails.

6.3 Usefulness of Uncertainty Maps

This subsection evaluates usefulness using the definition in Section 3.7. We measure usefulness by the Spearman rank correlation between pixelwise absolute error of the sample mean, $|\hat{\mu}_j - x^\star_j|$, and pixelwise variance, $\hat{v}_j$. A larger correlation indicates that pixels with larger error tend to have larger predicted variance.

Table 3: Usefulness measured by Spearman rank correlation between pixelwise absolute error $|\hat{\mu}_j - x_j^\star|$ and pixelwise variance $\hat{v}_j$. Results are reported as mean $\pm$ standard deviation over $N = 200$ images.

Method	Setting	Spearman ρ_{sp}
DDRM	$s = 4,\ \sigma_0 = 0.0$	0.520 ± 0.145
DDRM	$s = 4,\ \sigma_0 = 0.05$	0.455 ± 0.141
DDRM	$s = 8,\ \sigma_0 = 0.0$	0.477 ± 0.164
DDRM	$s = 8,\ \sigma_0 = 0.05$	0.426 ± 0.154
FPS-SMC ($M = 8$)	$s = 4,\ \sigma_0 = 0.05$	0.292 ± 0.140
FPS-SMC ($M = 20$)	$s = 4,\ \sigma_0 = 0.05$	0.284 ± 0.138
FPS-SMC ($M = 8$)	$s = 8,\ \sigma_0 = 0.05$	0.286 ± 0.144
FPS-SMC ($M = 20$)	$s = 8,\ \sigma_0 = 0.05$	0.280 ± 0.142

Table 3 reports usefulness across settings. For DDRM, the Spearman correlation is positive in all four settings and ranges from 0.426 to 0.520. This indicates that the variance map $\hat{v}$ carries information about where errors occur. The correlation decreases under measurement noise and at the larger scale factor. This is consistent with the inverse problem becoming harder, so that the variance map is a weaker predictor of error magnitude.

For FPS-SMC, the correlation is also positive but smaller, and lies around 0.28 to 0.29 under both particle budgets. This indicates weaker rank alignment between $|\hat{\mu}_j - x_j^\star|$ and $\hat{v}_j$ in these experiments. As noted in Section 6.2, usefulness should be interpreted together with calibration. A method can rank error prone pixels correctly while still producing intervals that are too narrow.

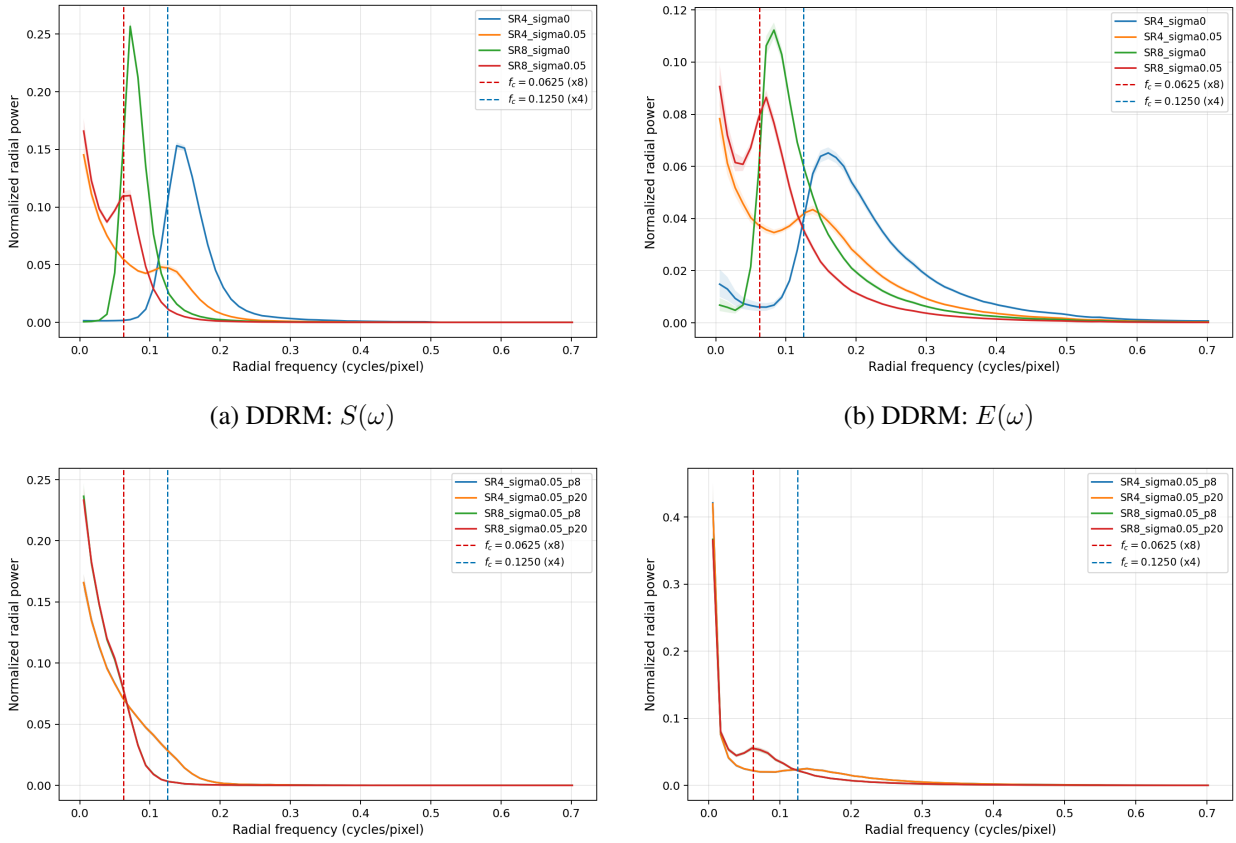
Comparing solvers in the noisy settings, DDRM achieves higher Spearman correlation than FPS-SMC. A plausible explanation is that DDRM variance is more spatially concentrated around high gradient structures, where errors often occur. This behaviour is illustrated in Figure 3, where DDRM variance is concentrated near object boundaries and fine edges. In contrast, FPS-SMC exhibits broader variance that is also visible in regions that appear well constrained by the LR observation. This broader pattern may reflect a different allocation of uncertainty, but it does not necessarily improve rank alignment with the error map. In addition, FPS-SMC uses sequential Monte Carlo resampling, which can reduce sample diversity under limited particle budgets. This can weaken the variance signal and may contribute to lower usefulness scores. These explanations are consistent with the calibration results at $s = 8$, where FPS-SMC shows strong under-coverage without a corresponding increase in interval width.

6.4 Frequency Diagnostics

This subsection evaluates the frequency diagnostics defined in Section 3. The goal is to test whether uncertainty is allocated to frequency bands where reconstruction error occurs. We report the normalised radial residual uncertainty spectrum $S(\omega)$ and the normalised radial error spectrum $E(\omega)$. We also report three scalar ratios computed using the cutoff $f_c = 0.5/s$ on the high resolution grid. These are the uncertainty ratio R_U, the error ratio R_E, and the under-exploration ratio $R_{UE} = R_U/R_E$.

Table 4: Frequency ratio diagnostics on the luminance Y channel. R_U is the high-to-low frequency ratio of the residual uncertainty spectrum. R_E is the corresponding ratio for the error spectrum. $R_{UE} = R_U/R_E$ is the under exploration ratio. The cutoff is $f_c = 0.5/s$ as defined in Section 3. Results are reported as mean $\pm$ standard deviation over $N = 200$ images.

Method	Setting	R_U	R_E	R_{UE}
DDRM	$s = 4,\ \sigma_0 = 0$	9.164 ± 1.645	28.851 ± 15.789	0.585 ± 0.733
DDRM	$s = 4,\ \sigma_0 = 0.05$	0.974 ± 0.578	4.247 ± 2.214	0.266 ± 0.181
DDRM	$s = 8,\ \sigma_0 = 0$	12.222 ± 3.097	47.387 ± 25.520	0.329 ± 0.208
DDRM	$s = 8,\ \sigma_0 = 0.05$	1.897 ± 0.847	7.532 ± 3.573	0.289 ± 0.136
FPS-SMC ($M = 8$)	$s = 4,\ \sigma_0 = 0.05$	0.332 ± 0.194	2.981 ± 1.899	0.160 ± 0.215
FPS-SMC ($M = 20$)	$s = 4,\ \sigma_0 = 0.05$	0.333 ± 0.205	2.965 ± 1.889	0.161 ± 0.211
FPS-SMC ($M = 8$)	$s = 8,\ \sigma_0 = 0.05$	0.641 ± 0.328	4.648 ± 2.808	0.193 ± 0.174
FPS-SMC ($M = 20$)	$s = 8,\ \sigma_0 = 0.05$	0.634 ± 0.326	4.657 ± 2.813	0.190 ± 0.170



(a) DDRM: $S(\omega)$

(b) DDRM: $E(\omega)$

(c) FPS-SMC: $S(\omega)$

(d) FPS-SMC: $E(\omega)$

Figure 6: Normalised radial spectra on the luminance Y channel. $S(\omega)$ is the residual uncertainty spectrum and $E(\omega)$ is the error spectrum (Section 3). Curves are normalised to unit energy. Vertical dashed lines indicate the cutoff $f_c = 0.5/s$ used for the HF/LF split ($f_c = 0.125$ for $s = 4$ and $f_c = 0.0625$ for $s = 8$). FPS-SMC is shown only for $\sigma_0 = 0.05$, and includes both particle budgets.

Fig. 6 shows two spectra of the solvers. For DDRM, the noise-free settings show a clear change in uncertainty allocation between $s = 4$ and $s = 8$. In both cases, we have uncertainty concentrating around the boundary frequency implied by the LR measurement. When $\sigma_0 = 0.05$, the normalised uncertainty spectrum shifts towards lower frequencies. This is consistent with extra uncertainty that is induced by the components that are observable but corrupted by noise. The error spectrum also shifts towards lower frequencies under noise. However, the under-coverage reported in Section 6.2 suggests that the uncertainty magnitude is still insufficient.

For FPS-SMC, both the residual uncertainty spectrum and the error spectrum are concentrated at low frequencies in the noisy settings. This is consistent with the qualitative example in Section 3.1, where FPS-SMC shows variance over broad regions, including areas that appear well constrained by the LR observation. It is also consistent with Section 6.3, since variation in low-frequency regions can be less aligned with corresponding pixel errors.

Comparing $s = 4$ and $s = 8$, the FPS-SMC error spectrum places more relative energy in mid frequencies around $\omega \approx 0.1$. The corresponding uncertainty spectrum does not show a matching increase in this region. This suggests that error increases in these frequencies without a proportional increase in predicted uncertainty, which is also consistent with the stronger under-coverage observed at $s = 8$ for FPS-SMC in Section 6.2.

We can see from Table 4 that R_{UE} is below 1 for all settings, which indicates that uncertainty assigns less relative energy to high frequencies than the error does. For DDRM, R_U is large when $\sigma_0 = 0$ and drops sharply when $\sigma_0 = 0.05$. This indicates that adding measurement noise increases the relative contribution of low-frequency uncertainty, compared with the noise-free setting, where uncertainty is concentrated near the cutoff. For FPS-SMC, R_U is smaller than for DDRM in the noisy settings. The under-exploration ratio R_{UE} is also smaller. This means that, relative to the error spectrum, FPS-SMC allocates less uncertainty to high frequencies under the tested particle budgets.

The frequency diagnostics complement the spatial uncertainty maps in Section 3.1. A shift of $S(\omega)$ towards lower frequencies does not imply that uncertainty is absent from boundaries. Boundaries contain energy across many frequencies. In addition, the spectra are normalised to unit energy, so increased low-frequency uncertainty can reduce the relatively high-frequency share. In the noisy settings, DDRM has a larger R_{UE} than FPS-SMC, which shows that DDRM assigns more uncertainty to high frequencies relative to its error than FPS-SMC does, which is consistent with the edge concentrated variance patterns in Figure 3. However, $R_{UE} < 1$ for both solvers, so that high-frequency uncertainty remains under-explored relative to high-frequency error in our experiments.

7 Conclusion & Limitations

In this paper, we evaluated whether distributions induced by diffusion-based solvers can behave like calibrated Bayesian posterior samplers for bicubic SR. We examined models like DDRM and FPS-SMC under four settings defined by the scale factor $s \in \{4, 8\}$ and the noise level $\sigma_0 \in \{0, 0.05\}$. Throughout the study, we followed the benchmarking framework described in Section 3.

The results show an evident difference between point accuracy and uncertainty reliability, with DDRM achieving better accuracy performance on the sample mean under the same compute budget. However, both solvers turned out to exhibit strong marginal under-coverage. For DDRM, interval widths increase as the setting becomes harder, but this is not sufficient to achieve calibration. For FPS-SMC, quantile and Gaussian widths at $\gamma = 0.95$ are similar, which is consistent with approximately Gaussian pixelwise marginals at this nominal level. However, the mean widths change only slightly from $s = 4$ to $s = 8$, while coverage significantly decreases at $s = 8$. This means that reconstruction errors increase without a corresponding increase in predicted uncertainty.

Despite miscalibration, uncertainty remains informative in a weaker sense, as the Spearman correlation between pixelwise absolute error and variance is positive for both solvers. DDRM attains higher correlation than FPS-SMC in the noisy settings, which suggests stronger rank alignment under this metric. Frequency tests also support the conclusion that high-frequency uncertainty is limited. Across settings, we have $R_{UE} < 1$, so high-frequency uncertainty is underrepresented relative to high-frequency error.

Overall, these findings support the accuracy trap phenomenon defined in [14], which states that point estimate accuracy alone does not validate uncertainty. In the tested SR settings, DDRM and FPS-SMC produce uncertainty maps that can help locate likely errors, but they do not provide calibrated 95% intervals. Future work should test larger reverse step budgets, larger particle numbers, and additional forward operators, and should include diagnostics that probe joint spatial uncertainty.

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A Additional evaluation algorithms

This appendix provides pseudocode for two evaluation procedures described in Section 3. The algorithms operate on a fixed measurement y and its associated solver sample set.

Algorithm 2 Coverage and interval width from a sample set

Inputs: Samples $\mathcal{S} = \{x^{(k)}\}_{k=1}^{K'}$ for fixed y, ground truth image x^*, nominal levels Γ, interval type (quantile or Gaussian)
Outputs: Coverage values $\{\widehat{\mathrm{Cov}}(\gamma)\}_{\gamma\in\Gamma}$ and widths $\{W(\gamma)\}_{\gamma\in\Gamma}$
Flatten image entries into indices $j \in \{1, \dots, P\}$
Let $x_j^{(k)}$ denote entry j of sample $x^{(k)}$, and let x_j^* denote entry j of x^*
for all $\gamma \in \Gamma$ **do**
  $c \leftarrow 0$, $w \leftarrow 0$
  for $j = 1$ to P **do**
    Construct interval $I_j(\gamma) = [\ell_j(\gamma), u_j(\gamma)]$ from $\{x_j^{(k)}\}_{k=1}^{K'}$
    $c \leftarrow c + \mathbb{I}\big(x_j^* \in I_j(\gamma)\big)$
    $w \leftarrow w + (u_j(\gamma) - \ell_j(\gamma))$
  end for
  $\widehat{\mathrm{Cov}}(\gamma) \leftarrow c/P$
  $W(\gamma) \leftarrow w/P$
end for

Algorithm 3 Frequency diagnostics for SR uncertainty

Inputs: Samples $\mathcal{S} = \{x^{(k)}\}_{k=1}^{K'}$, sample mean $\hat{\mu}$, ground truth image x^*, scale factor s
Outputs: Normalised spectra $\bar{S}(\omega)$ and $\bar{E}(\omega)$, ratios R_{U}, R_{E}, and R_{UE}
Residuals $r^{(k)} \leftarrow x^{(k)} - \hat{\mu}$ for $k = 1, \dots, K'$
$S(\omega) \leftarrow \frac{1}{K'} \sum_{k=1}^{K'} \left|\mathcal{F}\big(r^{(k)}\big)(\omega)\right|^2$
Error image $e \leftarrow \hat{\mu} - x^*$
$E(\omega) \leftarrow |\mathcal{F}(e)(\omega)|^2$
Normalise: $\bar{S}(\omega) \leftarrow S(\omega) \big/ \sum_{\omega'} S(\omega')$, $\bar{E}(\omega) \leftarrow E(\omega) \big/ \sum_{\omega'} E(\omega')$
Cutoff $f_c \leftarrow 0.5/s$ (cycles per high resolution pixel)
Define $\Omega_{\mathrm{L}} = \{\omega : |\omega| < f_c\}$ and $\Omega_{\mathrm{H}} = \Omega \setminus \Omega_{\mathrm{L}}$
$R_{\mathrm{U}} \leftarrow \dfrac{\sum_{\omega\in\Omega_{\mathrm{H}}} \bar{S}(\omega)}{\sum_{\omega\in\Omega_{\mathrm{L}}} \bar{S}(\omega)}$
$R_{\mathrm{E}} \leftarrow \dfrac{\sum_{\omega\in\Omega_{\mathrm{H}}} \bar{E}(\omega)}{\sum_{\omega\in\Omega_{\mathrm{L}}} \bar{E}(\omega)}$
$R_{\mathrm{UE}} \leftarrow R_{\mathrm{U}}/R_{\mathrm{E}}$

A.1 Variance convergence check

We choose the sampling budgets K' using an empirical convergence check on pixelwise variance estimates. The check is performed in the hardest setting ($s = 8$, $\sigma_0 = 0.05$). For a small set of pixels, we compute the running unbiased sample variance as the pooled sample count k increases from 1 to K'. Each curve in Figures 7–9 corresponds to one selected pixel. Stabilisation of these curves suggests that the variance estimate is no longer changing substantially with additional samples. Based on these plots, DDRM appears to stabilise by $K' = 150$, while FPS-SMC stabilises by around $K' = 800$ for both particle budgets. We therefore use $K' = 150$ for DDRM and $K' = 800$ for FPS-SMC in the main experiments.

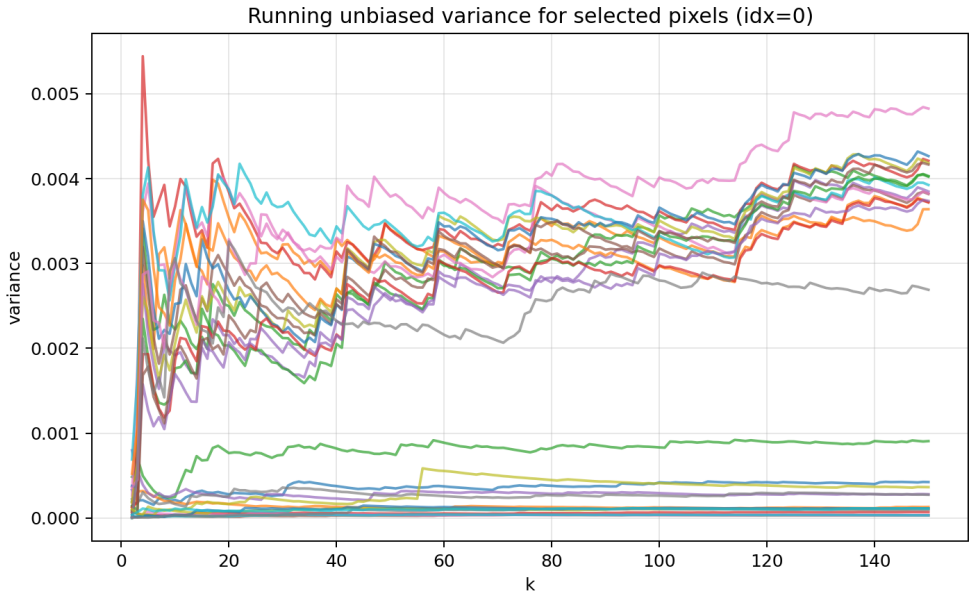


Figure 7: Running unbiased variance estimates for a set of selected pixels in the hardest setting ($s = 8$, $\sigma_0 = 0.05$) using DDRM. The horizontal axis is the pooled sample count k. Curves stabilise by around $k = 150$, which motivates using $K' = 150$ for DDRM.

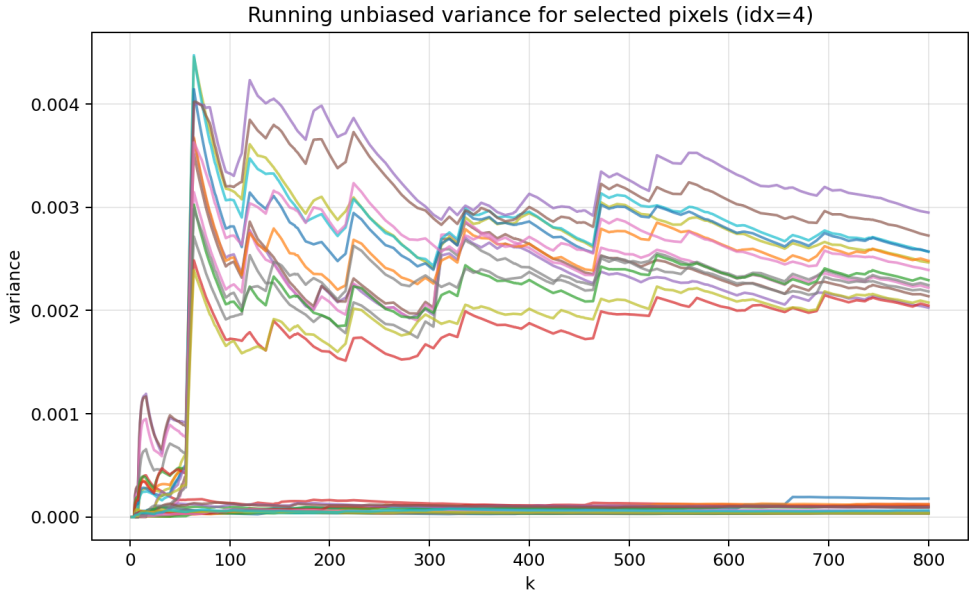


Figure 8: Running unbiased variance estimates for selected pixels in the hardest setting ($s = 8$, $\sigma_0 = 0.05$) using FPS-SMC with $M = 8$. The horizontal axis is the pooled sample count k. Curves stabilise by around $k = 700$, which motivates using $K' = 800$ for FPS-SMC to ensure convergence.

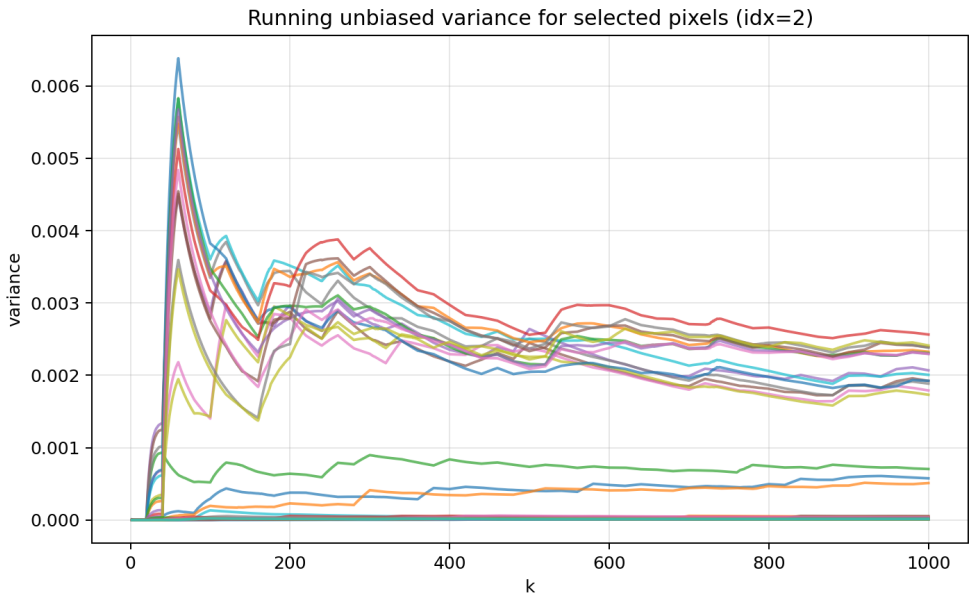


Figure 9: Running unbiased variance estimates for selected pixels in the hardest setting ($s = 8$, $\sigma_0 = 0.05$) using FPS-SMC with $M = 20$. The horizontal axis is the pooled sample count k. Curves show similar stabilisation behaviour to the $M = 8$ case, so we also use $K' = 800$ for this particle budget.

B Additional qualitative examples

This appendix provides qualitative examples to support the interpretation of the uncertainty metrics in Section 6. All examples use a fixed measurement y for repeated runs. Within each figure, the error and variance maps use shared colour scales across solvers. Sample visualisations use fixed indices within the pooled sample set to avoid cherry picking.

B.1 Noisy SR8 example on the same image as Section 3.1

Figure 10 shows SR8 with $\sigma_0 = 0.05$ for the same ground truth image used in Section 3.1. Both solvers produce similar sample mean reconstructions at a coarse scale. However, the variance maps differ in spatial allocation. DDRM variance is concentrated around boundaries and thin structures. FPS-SMC variance is more spatially spread and remains visible in smoother regions. This figure is shown for illustrative purposes and does not establish which solver is closer to the true posterior.

B.2 Additional qualitative examples for noise free settings

This appendix provides qualitative examples for $\sigma_0 = 0$ to illustrate spatial uncertainty patterns when uncertainty is dominated by ambiguity from downsampling rather than by measurement noise. In these examples, variance is concentrated near boundaries and thin structures and is nearly invisible in smoother regions.

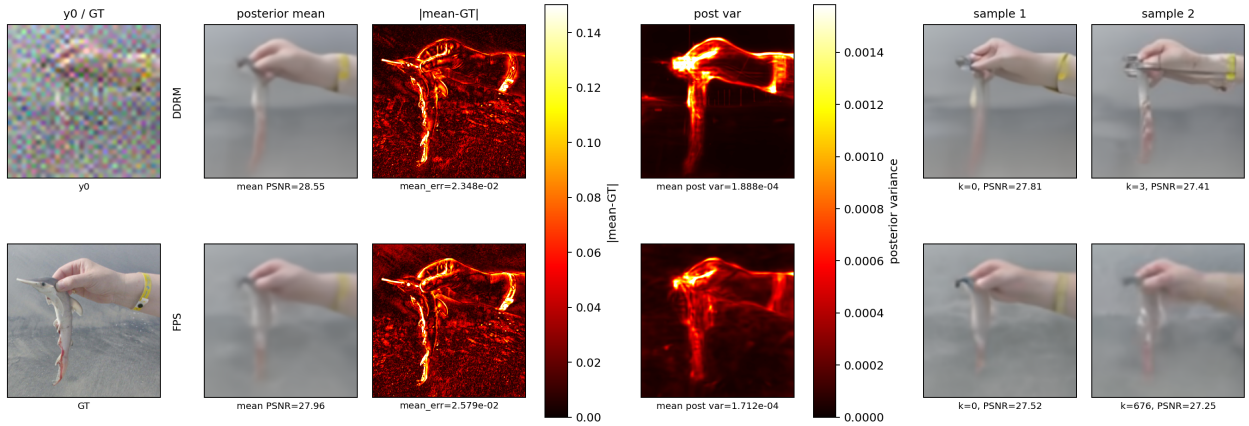


Figure 10: Qualitative comparison for SR8 with $\sigma_0 = 0.05$ on a fixed measurement y for the same image used in Section 3.1. Rows correspond to solvers (DDRM on top, FPS-SMC on bottom). Columns show the measurement visualisation, the sample mean $\hat{\mu}$, the absolute error map $|\hat{\mu} - x^*|$, the variance map $\hat{v}$, and two posterior samples. Error and variance maps use shared colour scales across solvers. Samples are selected at fixed indices from the pooled sample set for each solver.

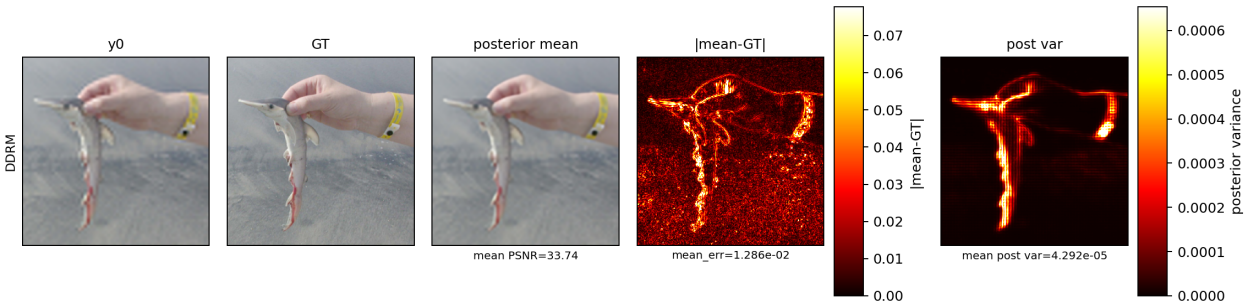


Figure 11: Noise free qualitative example for DDRM with $s = 4$ and $\sigma_0 = 0$. The figure shows the measurement visualisation y, the ground truth x^*, the sample mean $\hat{\mu}$, the absolute error map $|\hat{\mu} - x^*|$, and the variance map $\hat{v}$. Variance is concentrated around boundaries and fine structures.

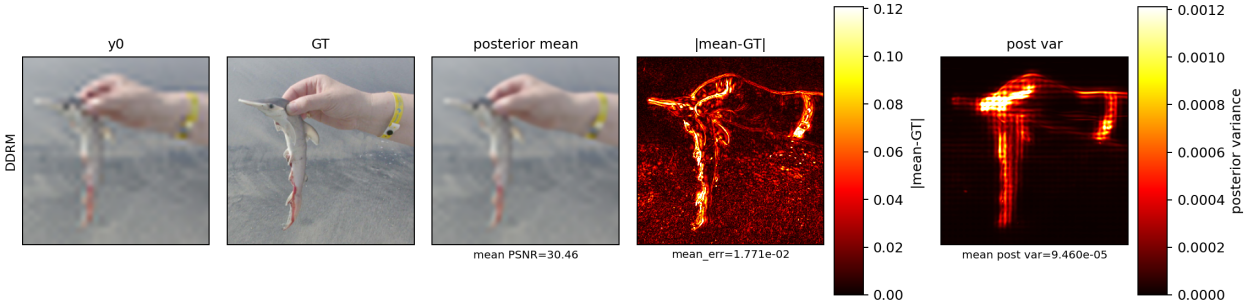


Figure 12: Noise free qualitative example for DDRM with $s = 8$ and $\sigma_0 = 0$ on the same image as Figure 11. Columns match Figure 11. Variance remains concentrated near boundaries and thin structures in this setting.